1 (a) $P = \frac{RT}{V-b} - \frac{a}{V^2}$ ← van der Waals 力 = 引力

↑ 斥力

$$dP = \left(\frac{\partial P}{\partial T}\right)_V dT + \left(\frac{\partial P}{\partial V}\right)_T dV$$

$$= \frac{\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P}{-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T} dT - \frac{1}{-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T} \cdot \frac{dV}{V}$$

$$= \frac{\alpha}{k} dT - \frac{1}{k} \cdot \frac{dV}{V}$$

$$dp = \left\{\frac{R}{V-b}\right\} dT + \left\{-\frac{RT}{(V-b)^2} + \frac{2a}{V^3}\right\} dV$$

$$\frac{\alpha}{k} = \frac{R}{V-b}, \quad \frac{1}{kV} = \frac{RT}{(V-b)^2} - \frac{2a}{V^3}$$ ⌋

(b) $\left(\frac{\partial V}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_P = -1$

$$\frac{-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T}{\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P} \cdot \left(\frac{\partial P}{\partial T}\right)_V = 1 \quad \rightarrow \quad \frac{k}{\alpha}\left(\frac{\partial P}{\partial T}\right)_V = 1$$ ⌋

1 (c) $\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p = A(1 - \eta p)$

$$K = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T = k(1 + \beta(T - T_0))$$

$$\frac{\alpha}{K} = \left(\frac{\partial P}{\partial T}\right)_V$$

$$dp = \frac{\alpha(p)}{k(T)}\,dT - \frac{1}{k(T)} \cdot \frac{dV}{V}$$

$\eta A = k\beta = C$

$$\rightarrow \frac{dV}{V} = \alpha(p)\,dT - K(T)\,dp$$

$d(\alpha K) = \alpha\,dK + K\,d\alpha$

$d(\alpha K)$

$\ln V$

$x\,dy + y\,dx$

$\ln V = \frac{1}{C}\alpha K$

$d(\alpha(p)\,K(T))$

$= \alpha(p)\,K'\,dT + K\,\alpha'\,dp$

$k\beta \qquad -A\alpha$

2. $p = f(V)T = \frac{R}{V}T$

$$dS = \frac{1}{T}dU + \frac{p}{T}dV$$

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{1}{T}\left(\frac{\partial U}{\partial V}\right)_T + \frac{p}{T}$$

$$\left(\frac{\partial S}{\partial T}\right)_V = \frac{1}{T}\left(\frac{\partial U}{\partial T}\right)_V \quad \leftarrow \frac{dS}{dT} = \frac{1}{T}\frac{dU}{dT} + \frac{p}{T}\frac{dV}{dT}^{\to 0} \quad (V = \text{const})$$

dS が完全微分 $\Leftrightarrow \frac{\partial^2 S}{\partial T \partial V} = \frac{\partial^2 S}{\partial V \partial T}$

$$\frac{\partial}{\partial T}\left(\frac{1}{T}\frac{\partial U}{\partial V}\right) + \frac{\partial}{\partial T}\left(\frac{p}{T}\right) = \frac{\partial}{\partial V}\left(\frac{1}{T}\frac{\partial U}{\partial T}\right)$$

$$\left(\frac{\partial U}{\partial V}\right) - T^2\frac{\partial}{\partial T}\left(\frac{p}{T}\right) = 0$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T^2\frac{\partial}{\partial T}\left(\frac{p}{T}\right)_V = 0$$

3. (a) $\left.\begin{aligned} d'Q &= T\,dS \\ d'Q &= dU + P\,dV \end{aligned}\right\} \rightarrow dS = \frac{1}{T}dU + \frac{P}{T}dV$

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{1}{T}\left(\frac{\partial U}{\partial V}\right)_T + \frac{P}{T}$$

$$\left(\frac{\partial S}{\partial T}\right)_V = \frac{1}{T}\left(\frac{\partial U}{\partial T}\right)_V$$

dS が完全微分 $\Leftrightarrow \frac{\partial^2 S}{\partial T\,\partial V} = \frac{\partial^2 S}{\partial V\,\partial T}$

$$\frac{\partial}{\partial T}\left(\frac{1}{T}\frac{\partial U}{\partial V}\right) + \frac{\partial}{\partial T}\left(\frac{P}{T}\right) = \frac{\partial}{\partial V}\left(\frac{1}{T}\frac{\partial U}{\partial T}\right)$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T^2\frac{\partial}{\partial T}\left(\frac{P}{T}\right)_V$$

$$= T\left(\frac{\partial P}{\partial T}\right)_V - P \quad \lrcorner$$

3 (b) $P = \dfrac{RT}{V-b} - \dfrac{a}{V^2}$

$C_V = \left(\dfrac{\partial U}{\partial T}\right)_V \rightarrow \left(\dfrac{\partial C_V}{\partial V}\right)_T = \dfrac{\partial^2 U}{\partial V \partial T} = \dfrac{\partial^2 U}{\partial T \partial V}$

$= \dfrac{\partial}{\partial T}\left\{ T\left(\dfrac{\partial P}{\partial T}\right)_V - P \right\}$ ← $\left(\dfrac{\partial U}{\partial V}\right)_T$

この状態量で決定される.

$= T\left(\dfrac{\partial^2 P}{\partial T^2}\right)_V = 0$

$dU = \left(\dfrac{\partial U}{\partial T}\right)_V dT + \left(\dfrac{\partial U}{\partial V}\right)_T dV = C_V dT + \left\{ T\left(\dfrac{\partial P}{\partial T}\right)_V - P \right\} dV$

$= C_V dT + \dfrac{a}{V^2} dV$

$U = \int^T C_V dT - \dfrac{a}{V} + U_0$

$dS = \dfrac{1}{T} dU + \dfrac{P}{T} dV = \dfrac{C_V}{T} dT + \dfrac{R}{V-b} dV$

$S = \int^T \dfrac{C_V}{T} dT + R \ln(V-b) + S_0$

4. 張力 $f = aT\left\{\frac{l}{l_0} - \left(\frac{l}{l_0}\right)^{-2}\right\}$

$$\begin{cases} dU = TdS + f dl \\ dF = -SdT + f dl \end{cases}$$

(a) $f = -\frac{\partial U}{\partial l} \rightarrow U = -\int_0^{l_0} f\, dl$

$\left(\frac{\partial U}{\partial l}\right)_T = 0$

$$= -aT l_0 \int_0^1 \left(x - \frac{1}{x^2}\right) dx$$

$$= -aT l_0 \left[\frac{x^2}{2} + \frac{1}{x}\right]_0^1$$

(b) $f = -T\left(\frac{\partial S}{\partial l}\right)_T$

$d'Q = TdS = T\left(\frac{\partial S}{\partial l}\right)_T dl = -f dl \quad (dT = 0)$

(c) $Q = \int_{l_0}^{l} (-f)\, dl$

$= aT\left(\frac{1}{2}\frac{l^2}{l_0} + \frac{l_0^2}{l} - \frac{3}{2} l_0\right)$

(d) $dS = 0 \rightarrow dT = \left(\frac{\partial T}{\partial l}\right)_S dl = \frac{f}{C_l} dl \qquad \left(\frac{\partial S}{\partial T}\right)_l = \frac{C_l}{T}$

$$\ln \frac{T}{T_0} = \frac{a}{C_l}\left(\frac{1}{2}\cdot\frac{l^2}{l_0} + \frac{l_0^2}{l} - \frac{3}{2} l_0\right)$$

5. $C_x = \left(\frac{d'Q}{dT}\right)_x = T\left(\frac{dS}{dT}\right)_x$

(a) $$\frac{C_v}{C_p}\left(\frac{\partial P}{\partial V}\right)_S = \frac{\dfrac{\partial(S,V)}{\partial(T,V)}}{\dfrac{\partial(S,P)}{\partial(T,P)}} \cdot \frac{\partial(P,S)}{\partial(V,S)} = \frac{\partial(T,P)}{\partial(T,V)} = \left(\frac{\partial P}{\partial V}\right)_T$$

(b) $$\frac{\partial^2 U}{\partial V^2} = \frac{\partial}{\partial V}\left(\frac{\partial U}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial V}\right)_S$$

$$\frac{\partial^2 U}{\partial S^2} = \frac{\partial}{\partial S}\left(\frac{\partial U}{\partial S}\right)_V = \left(\frac{\partial T}{\partial S}\right)_V$$

$$\left(\frac{\partial^2 U}{\partial V \partial S}\right)^2 = \frac{\partial}{\partial V}\left(\frac{\partial U}{\partial S}\right)_V \cdot \frac{\partial}{\partial S}\left(\frac{\partial U}{\partial V}\right)_S = -\left(\frac{\partial T}{\partial V}\right)_S \left(\frac{\partial P}{\partial S}\right)_V$$

$G = H - TS$ $\qquad$ $H = U + pV$ （ヘルムホルツポテンシャル）

$dG = dH - TdS - SdT$ $\qquad$ $dH = dU + pdV + Vdp = TdS + Vdp$

$= Vdp - SdT$ $\qquad$ $dQ = TdS$

化学平衡では等温なので

$dG = Vdp$

$$G = G_0 + \int Vdp = G_0 + \int \frac{RT}{P} dp = G_0 + RT \ln P$$

（G_0：1atm, 25℃）

$$mA \rightleftharpoons mB$$

$$\left.\begin{aligned} G_A &= mG_A^\circ + mRT \ln P_A \\ G_B &= mG_B^\circ + mRT \ln P_B \end{aligned}\right\}$$ 平衡の条件 $G_A = G_B$

（化学反応は $\Delta G \leq 0$ <0 不可逆反応 $=0$ 可逆反応）

$$mG_A^\circ - mG_B^\circ = RT \ln \frac{P_B^m}{P_A^n}$$

（$mG_A^\circ - mG_B^\circ = \Delta G^\circ_{反応}$） $\qquad$ K_p 圧平衡定数

$$K_p = e^{\frac{\Delta G^\circ}{RT}}$$

$\Delta G^\circ_{反応} = -\Delta H + T\Delta S$ （反応熱 Q） $\qquad \rightarrow K_p = e^{\frac{\Delta S}{R}} \cdot e^{\frac{Q}{RT}}$

$mA = mB + Q$



$\epsilon \ll 1$, $P_A = \text{const}$, $P_B = \text{const}$

$$W_a = P_A \cdot (n\epsilon V_A) - P_B (m\epsilon V_B) = (n-m)\epsilon RT$$

$$G = H - T(S + \Delta S) = \sum n_i G_i^\circ - RT \sum n_i \ln x_i \qquad \Delta S = -R \sum n_i \ln x_i$$

$$= \sum n_i [G_i^\circ + RT \ln x_i]$$

$$\delta T = 0 \; ; \; \delta G = \sum [G_i^\circ + RT \ln x_i] \delta n_i + RT \sum n_i \delta(\ln x_i) = 0$$

$$\sum n_i \frac{\delta x_i}{x_i} = \sum \delta x_i \sum n_i = 0$$

$$\delta n_i \propto \eta_i$$

$$\sum [G_i^\circ + RT \ln x_i] \eta_i = 0$$

$$\sum \eta_i \ln x_i = -\frac{1}{RT} \sum \eta_i G_i^\circ$$

$$\ln K$$

6 (1) $dU = -pdV$ より $H = U + PV = \text{const}$

$dH = 0 \rightarrow dS = -\frac{V}{T}dp$

(2) $0 = \int_1^2 \frac{d'Q}{T} < S_2 - S_1 = \int_{p_1}^{p_2} -\frac{V}{T}dp > 0 \qquad p_1 > p_2$

(3) $U_2 + P_2V_2 - (U_1 + P_1V_1) = U(V_2, T_2) - U(V_1, T_1) + RT_2 - RT_1$

$= \left(\frac{\partial U}{\partial V}\right)_T (\exists V, T_2)(V_2 - V_1) + \left(\frac{\partial U}{\partial T}\right)_V (V_1, \exists T)(T_2 - T_1) + R(T_2 - T_1)$

$U = \frac{3}{2}nRT$

$= \left\{\left(\frac{\partial U}{\partial T}\right)_V + R\right\}(T_2 - T_1) = 0$

($\left(\frac{\partial U}{\partial V}\right)_T = 0$, $\left(\frac{\partial U}{\partial T}\right)_V = 0$)

$dH = 0 \rightarrow \left(\frac{\partial T}{\partial P}\right)_H = \frac{1}{C_p}\left\{-V + T\left(\frac{\partial V}{\partial T}\right)_p\right\}$

$\simeq \frac{nb}{C_pT}\left(\frac{2a}{Rb} - T\right) \qquad (a, b \ll 1)$

$T < \frac{2a}{Rb}$ のとき $dp < 0$ より $dT < 0$

$T > \frac{2a}{Rb}$ $dT > 0$

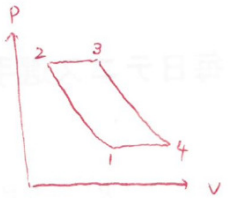
(4) $U = \int C_v dT - \frac{a}{V} \qquad C_v = C_v(T)$ cf. 3(b)

$S = \int \frac{C_v}{T}dT + R\ln(V - b)$

$\begin{cases} dS = \frac{R}{V-b}dV + \frac{C_v}{T}dT = \frac{P + \frac{a}{V^2}}{T} \cdot (-\kappa V dp + \alpha dT)V \\ dS = -\frac{V}{T}dp \end{cases}$

7. a. $u = f(T)$

b. $p = \frac{1}{3}u$



1→2 断熱 $dU + pdV = 0$

$u = AV^{-\frac{4}{3}}$

2→3 等温 $dU = (V_3 - V_2)u_2$

$dW = \frac{1}{3}u_2(V_3 - V_2)$

→ 吸熱 $\frac{4}{3}u_1(V_3 - V_2)$

(1) $dU = TdS - pdV$

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial S}{\partial V}\right)_T - p = T\left(\frac{\partial p}{\partial T}\right)_V - p$$

$$U = Vu \rightarrow u = T\frac{1}{3}\frac{du}{dT} - \frac{1}{3}u \rightarrow \frac{du}{u} = 4\frac{dT}{T}$$

Carnot Cycle $\frac{Q_1}{T_1} = \frac{Q_2}{T_2} \rightarrow \frac{u_2}{T_2}\left\{\frac{V_1}{V_1}\left(\frac{u_1}{u_2}\right)^{\frac{3}{4}}\right\} = \frac{u_1}{T_1}\left\{\frac{V_2}{V_1}\left(\frac{u_2}{u_1}\right)^{\frac{3}{4}} - 1\right\} \rightarrow \frac{u^{\frac{1}{4}}}{T} = \text{const}$

(2) $LV_s = \text{const} \rightarrow V_s^3 V = \text{const} \rightarrow 3\frac{\delta V_s}{V_s} + \frac{\delta V}{V} = 0$

断熱 → $d(Vu) = -pdV$

$$\frac{4}{3}\frac{dV}{V} + \frac{du}{u} = 0$$

$$-4\frac{dV}{V} + 4\frac{dT}{T} = 0 \rightarrow \frac{V}{T} = \text{const}$$

(3) $\frac{U_s}{V_s} = \frac{V u_s}{V_s} = a\frac{T^4}{V_s^4}$ より $f(x) = \frac{a}{x^4}$

$\frac{E}{V} = \frac{U_s}{V_s}$ は断熱不変量, $\frac{U_s}{V_s} = f\left(\frac{V_s}{T}\right)$

$$U(\nu)d\nu = Z(\nu)U_s d\nu = \frac{8\pi\nu^2 V}{c^3} \cdot \nu f\left(\frac{\nu}{T}\right)d\nu$$

$$\frac{dU(\nu)}{d\nu} = 3\nu^2\frac{8\pi V}{c^3}f\left(\frac{\nu}{T}\right) + \frac{8\pi\nu^3 V}{c^3}\cdot\frac{1}{T}f'\left(\frac{\nu}{T}\right) \qquad \frac{d(U(\nu)}{d\nu} = 0 \text{ となる } \nu_m \text{ で}$$

$3f\left(\frac{\nu_m}{T}\right) + \frac{\nu_m}{T}f'\left(\frac{\nu_m}{T}\right) = 0$ $\frac{T}{\nu_m} = \text{const}$ は ことを示す.

8.

$$TdS = T\left(\frac{\partial S}{\partial T}\right)_H dT + T\left(\frac{\partial S}{\partial H}\right)_T dH$$

$$= C_H dT + T\left(\frac{\partial M}{\partial T}\right)_H dH \qquad (\because\ dU = TdS + HdM)$$

斷熱 $\rightarrow dS = 0$; $C_H dT = \frac{CH}{T} dH$ $\quad (\because\ M = \frac{CH}{T})$

$$\left.\begin{array}{l} TdS = C_H dT - MdH \\ U = aT^4 \rightarrow 4aT^3 dT = TdS + HdM \end{array}\right\} \rightarrow MdH = HdM \quad (T = \text{const})$$

$$\int H dM = \int M dH = \frac{C}{T}\int_0^H H dH = \frac{CH^2}{2T}$$

$$4acT^2 dT = CdS + MdM$$

$$S = \frac{4}{3}aT^3 - \frac{M^2}{2C} = \frac{4}{3}aT^3 - \frac{CH^2}{2T^2}$$

斷熱 $\rightarrow$ $S = \text{const}$ $\quad H \rightarrow 0$; T 減少

9. (a) $dG = Vdp - SdT \rightarrow d(\Delta G) = \Delta V\, dp - (\Delta S)\, dT$

相平衡 $\Delta G = 0$ $\quad \frac{dP}{dT} = \frac{\Delta S}{\Delta V} = \frac{\Delta q}{T\Delta V}$



$S = -\left(\frac{\partial G}{\partial T}\right)_P$

$(\Delta q = T\Delta S)$



$TdS = T\left(\frac{\partial S}{\partial P}\right)_T dP + T\left(\frac{\partial S}{\partial T}\right)_P dT$

(b) $\frac{dP}{dT} = \frac{P\Delta q}{RT^2}$ $\quad (\Delta V \simeq \frac{RT}{P})$

$C = \frac{d'Q}{dT} = \frac{TdS}{dT} = T\left(\frac{\partial S}{\partial P}\right)_T \frac{P\Delta q}{RT^2} + T\left(\frac{\partial S}{\partial T}\right)_P = -T\left(\frac{\partial V}{\partial T}\right)_P \frac{P\Delta q}{RT^2} + T\left(\frac{\partial S}{\partial T}\right)_P = -\frac{\Delta q}{T} + C_P$

($\frac{R}{P}$)

$\Delta q = -RT \ln P \rightarrow \frac{\Delta q}{\Delta T} = -R \ln P$

(c) $\frac{d'Q}{dT} < 0$

(d) 

$dP = \Delta q\, dT$

$\frac{dP}{dT} = \frac{d(\ln P)}{dT} = \frac{\Delta q}{RT^2}$

10.

(1) $$W_b = \int_{P_B}^{P_B'} p\,dV = \int_{P_B}^{P_B'} -\frac{n\epsilon RT}{P}\,dp = -n\epsilon RT \ln\left(\frac{P_B'}{P_B}\right)$$

$$W_d = -n\epsilon RT \ln\left(\frac{P_A}{P_A'}\right)$$

$$W = W_b + W_d = \epsilon RT\left\{\ln\left(\frac{P_B}{P_B'}\right)^m + \ln\left(\frac{P_A'}{P_A}\right)^n\right\}$$

$$= \epsilon RT \ln \frac{P_A'^n \cdot P_B^m}{P_A^n \cdot P_B'^m} = 0$$

$$\frac{P_A^n}{P_B^m} = \frac{P_A'^n}{P_B'^m}$$

T = const

$$dU = d'Q - pdV + \mu dN$$

$$U = \frac{3}{2}NkT \qquad \mu = \left(\frac{\partial U}{\partial N}\right)_{F,V} = \frac{3}{2}kT$$

11

n次元球の体積 $\frac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2}+1)} r^n$

運動量 $\frac{1}{2m}(P_{1x}^2 + P_{2x}^2 + \cdots + P_{1y}^2 + \cdots + P_{1z}^2 + \cdots + P_{Nz}^2)$

$P_{1x}^2 + \cdots + P_{Nz}^2 \leq 2mE$

$\Omega(E) = \frac{(2mE\pi)^{\frac{3}{2}N}}{\Gamma(\frac{3}{2}N+1)} V^N$

$S(E) = N \times (\text{const})$ より 示量性

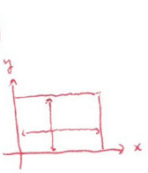
$C_n r^n = \int_{x_1^2+\cdots+x_n^2 \leq r^2} dx_1 dx_2 \cdots dx_n = \int_0^r dr\, n C_n r^{n-1}$

$\int_{-\infty}^{\infty} e^{-(x_1^2+\cdots+x_n^2)} dx_1 \cdots dx_n = \int_0^{\infty} n \cdot C_n \cdot r^{n-1} e^{-r^2} dr$

$\sqrt{\pi}^n = C_n \Gamma(\frac{n}{2}+1)$

12. $H = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)$

周期境界条件
$\psi(0)=\psi(L)$
$\psi'(0)=\psi'(L)$



$$\psi = \sum_{n_x n_y n_z} A \exp\left[2\pi i\left(\frac{n_x x}{L}+\frac{n_y y}{L}+\frac{n_z z}{L}\right)\right]$$

$$= A\exp\left[i(k_x x + k_y y + k_z z)\right] \qquad k = \frac{2\pi n}{L}$$

$$E = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2)$$

$$= \frac{\hbar^2}{2mL^2}(n_x^2 + n_y^2 + n_z^2) \qquad n = 0, \pm 1, \pm 2, \cdots$$

$g_L(E)\,dE$ = 「半径 $\sqrt{\frac{2mL^2(E+\Delta E)}{\hbar^2}}$ の球内の格子点の数」 − 「〃 $\sqrt{\frac{2mL^2E}{\hbar^2}}$ 〃」 } 格子間隔 $\frac{2\pi}{L}$

$$\simeq \frac{4\pi}{3}\left(\frac{2mL^2(E+\Delta E)}{\hbar^2}\right)^{\frac{3}{2}} - \frac{4\pi}{3}\left(\frac{2mL^2E}{\hbar^2}\right)^{\frac{3}{2}}$$

$$= \frac{4\pi}{3}\cdot\frac{V}{h^3}\left\{(2m(E+\Delta E))^{3/2} - (2mE)^{3/2}\right\}$$

$$\simeq 2\pi\,\frac{V}{h^3}(2m)^{3/2}\sqrt{E}\,\Delta E$$

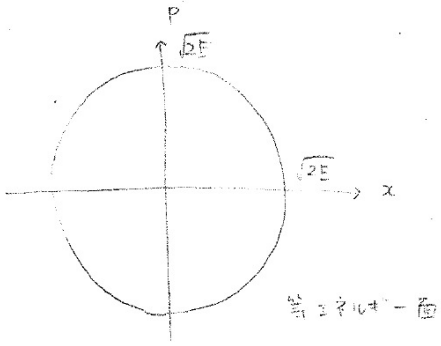
古典的位相体積

$$\int_{2mE < p^2 < 2m(E+\Delta E)} d^3p\,d^3\xi = V\cdot\int d^3p = V\cdot\frac{4\pi}{3}\left\{(2m(E+\Delta E))^{3/2} - (2mE)^{3/2}\right\}$$

$$\simeq 2\pi V(2m)^{3/2}\sqrt{E}\,\Delta E$$

運動量空間での単位セル h^3

13. $H = \frac{1}{2}p^2 + \frac{1}{2}x^2$



等エネルギー面

分布関数 $P(E) \propto e^{-\frac{E}{kT}}$

$$\langle E \rangle = \frac{\int E e^{-\frac{E}{kT}} dp dx}{\int e^{-\frac{E}{kT}} dp dx}$$

断面積原理

$$= \frac{\int E e^{-\frac{E}{kT}} dE}{\int e^{-\frac{E}{kT}} dE}$$

$\frac{1}{2}p^2 + \frac{1}{2}x^2 = E$

$dp\,d\theta = \sqrt{2E}\,d\theta\,d\sqrt{2E} = d\theta\,dE$

(1) $$\langle E \rangle = \frac{\left[-kT(E+kT)e^{-\frac{E}{kT}}\right]_0^\infty}{\left[-kT e^{-\frac{E}{kT}}\right]_0^\infty} = kT$$

(2) $$\langle E \rangle = \frac{\sum_{n=0}^{\infty} n\epsilon e^{-\frac{n\epsilon}{kT}}}{\sum_{n=0}^{\infty} e^{-\frac{n\epsilon}{kT}}} = \frac{\epsilon \frac{e^{-\frac{\epsilon}{kT}}}{(1-e^{-\frac{\epsilon}{kT}})^2}}{\frac{1}{1-e^{-\frac{\epsilon}{kT}}}} = \frac{\epsilon}{e^{\frac{\epsilon}{kT}} - 1} \longrightarrow kT$$

$e^{\frac{\epsilon}{kT}} \sim 1 + \frac{\epsilon}{kT}$

$$\sum e^{-\alpha n} = \frac{1}{1-e^{-\alpha}}$$

$$-\sum n e^{-\alpha n} = \frac{d}{d\alpha}\left(\frac{1}{1-e^{-\alpha}}\right)$$

14 (1) 1 particle $p = \frac{v}{V}$

m particle $P(n) = {}_N C_n \lambda^n (1-\lambda)^{N-n}$ $\lambda = \frac{v}{V}$

$$a = \langle n \rangle = N\lambda = N\frac{v}{V}$$

$$b = \langle (n - \langle n \rangle)^2 \rangle = N\lambda(1-\lambda) = N\frac{v}{V}\left(1 - \frac{v}{V}\right)$$

(2) $$F(x) = \sum_{n=0}^{N} p(n) x^n = \left\{(1-\lambda) + \lambda x\right\}^N$$

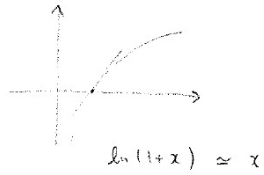
$x \sim 0$ で考える.

$$= \exp\left\{N \ln\left[(1-\lambda) + \lambda x\right]\right\}$$

$$= \exp\left\{N \ln\left[1 - \lambda(1-x)\right]\right\}$$

$$\simeq \exp\left[N\left\{-\lambda(1-x) + O(\lambda^2)\right\}\right]$$

$$\simeq e^{-a(1-x)}$$



$\lambda = \frac{v}{V} \to$ 一定値 $\to 0$ は物質が無い場合

$$P(n) = \frac{1}{n!}\left[\frac{d}{dx}\right]^n F(x)\Big|_{x=0} = \frac{a^n}{n!} e^{-a}$$

14. $\left(1-\frac{v}{V}\right)^{N-n} \sim \left(1-\frac{v}{V}\right)^{N} \sim \left(1-\frac{\rho v}{V}\right)^{N} = e^{-\rho v} = e^{-a}$

15. (1)

$$\ln P(n) \simeq N\ln N - (N-n)\ln(N-n) - n\ln n + n\ln\lambda + (N-n)\ln(1-\lambda)$$

$$\frac{d}{dn}\ln P(n^*) = -\ln\frac{n^*}{N-n^*} + \ln\frac{\lambda}{1-\lambda} = 0 \qquad n^* = N\lambda$$

$$\ln P(n) = \ln P(n^*) + \frac{1}{2}\left[\ln P(n^*)\right]''(n-n^*)^2 + \cdots \qquad O(N^{-m+1})\cdot(n-n^*)^m$$

$$N^{-2}(n-n^*)^3$$

$$= \ln P(n^*) - \frac{1}{2}\left(\frac{1}{n^*} + \frac{1}{N-n^*}\right)(n-n^*)^2 + \cdots \qquad N^{-m+1}(x\sqrt{N})^m = x^m N^{1-\frac{m}{2}}$$

$$\| \qquad\qquad \| \qquad\qquad x = \frac{n-n^*}{\sqrt{N\lambda(1-\lambda)}}$$

$$0 \qquad \frac{N}{n^*(N-n^*)} = \frac{1}{N\lambda(1-\lambda)}$$

$$= -\frac{1}{2}x^2 + x^m O(N^{1-\frac{m}{2}}) \quad (m\geq 3) \qquad x = \frac{\cdots}{\sqrt{N}}$$

$n-n^* \sim \sqrt{N}$ $\qquad O(1)^{-2}(n-n^*)^3$

$O(1^{-1}\cdot N^{\frac{3}{2}}) \qquad e^{\sqrt{N}x^3}$

$O(1)^{-3}\sqrt{N\lambda(1-\lambda)}\,x^3$

(2) $\quad dx = \frac{dn}{\sqrt{N\lambda(1-\lambda)}}$

$$P(n(x))\,dn = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx$$

16. (i) $E_i = s_i \hbar\omega$

$s_1 + s_2 + \cdots + s_N = n$

状態数 $g(N,n)$ は $s_1 \cdots s_N$ に 1 を重複を許して入れる組の数

$$g(N,n) = {}_{N+n-1}C_n = \frac{(N+n-1)!}{n!(N-1)!}$$

(ii) $$\frac{1}{T} = \frac{k}{\hbar\omega}\frac{\partial(\log g(N,n))}{\partial n}$$

$$\log g(N,n) \sim (N+n-1)(\log(N+n-1) - 1) - n(\log n - 1) - \log(N-1)!$$

$$\frac{N+n}{n} = e^{\frac{\hbar\omega}{kT}} \qquad n = \frac{N}{e^{\frac{\hbar\omega}{kT}} - 1}$$

$E = n\hbar\omega$

[13] (2) $$\frac{\sum_{n=0}^{\infty} n\epsilon e^{-\frac{n\epsilon}{kT}}}{\sum_{n=0}^{\infty} e^{-\frac{n\epsilon}{kT}}} = \frac{\epsilon}{e^{\frac{\epsilon}{kT}} - 1}$$

$$E = \frac{N\sum_{p=0}^{\infty} p\hbar\omega\, e^{-\frac{p\hbar\omega}{kT}}}{\sum_{p=0}^{\infty} e^{-\frac{p\hbar\omega}{kT}}}$$

17 (i) $\langle \xi_i \rangle = \dfrac{\int \xi_i e^{-\beta E} d\xi dp}{\int e^{-\beta E} d\xi dp} = \dfrac{\int \xi_i e^{-\sum \beta \xi_i(p_i)} e^{-\beta E'} d\xi dp}{\int e^{-\sum \beta \xi_i(p_i)} e^{-\beta E'} d\xi dp} = \dfrac{\int \xi_i e^{-\beta \xi_i(p_i)} dp_i \prod_{j \neq i} \int e^{-\beta \xi_j(p_j)} dp_j}{\prod_j \int e^{-\beta \xi_i(p_i)} dp_j}$

$= -\dfrac{\partial}{\partial \beta} \left(\ln \int e^{-\beta \xi_i} dp_i \right)$

(ii) $\overline{\xi_i} = -\dfrac{\partial}{\partial \beta} \left(\ln \int e^{-\beta b_i p_i^2} dp_i \right) = -\dfrac{\partial}{\partial \beta} \ln \sqrt{\dfrac{\pi}{\beta b_i}} = -\dfrac{\partial}{\partial \beta} \left(\ln \sqrt{\dfrac{\pi}{b_i}} - \dfrac{1}{2} \ln \beta \right) = \dfrac{1}{2\beta} = \dfrac{1}{2} kT$

(iii) $c_{ij} = c_{ji} = \dfrac{1}{2} b_{ij} \quad (i \neq j) \qquad c_{ii} = b_{ii}$

$\sum_{i \geq j} b_{ij} p_i p_j = \sum_i b_{ii} p_i p_i + \sum_{i>j} b_{ij} p_i p_j = \sum_{i,j} c_{ij} p_i p_j$

$C = (c_{ij})$ 対称

$$ {}^t T C T = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_f \end{pmatrix} \qquad \begin{pmatrix} p_1 \\ \vdots \end{pmatrix} = {}^t T \begin{pmatrix} P_1 \\ \vdots \end{pmatrix} \qquad \begin{pmatrix} Q_1 \\ \vdots \end{pmatrix} = {}^t T \begin{pmatrix} \xi_1 \\ \vdots \end{pmatrix} $$

$$ \sum p_r \dot{Q}_r = (P_1 \cdots P_f) \begin{pmatrix} \dot{Q}_1 \\ \vdots \\ \dot{Q}_f \end{pmatrix} = (P_1 \cdots P_f)\, T\, {}^t T \begin{pmatrix} \dot{\xi}_1 \\ \vdots \end{pmatrix} = \sum P_r \dot{\xi}_r $$

$H(\xi_1, \cdots, p_f) = H(Q_1, \cdots P_f) \qquad (H = \sum p_r \dot{\xi}_r - L)$

$\dot{Q}_i = \tau_{ij} \dot{\xi}_j = \tau_{ij} \dfrac{\partial H}{\partial p_j} = \dfrac{\partial H}{\partial P_i} \qquad \dot{P}_i = -\dfrac{\partial H}{\partial Q_i}$ P, Q は正準変数

$E = \sum a_i P_i^2 + E'(Q_1 \cdots Q_f)$ $\overline{a_i P_i^2}$ に対して エネルギー等分配則 成立

$\left\langle \sum_{i \geq j} b_{ij} p_i p_j \right\rangle = \dfrac{1}{h^f} \int d^f p\, d^f \xi \sum_{i \geq j} b_{ij} p_i p_j e^{-\frac{1}{kT} \left(\sum_{i \geq j} b_{ij} p_i p_j + E'(\xi) \right)}$

$= \dfrac{kT}{2} \int d^f p\, d^f \xi \dfrac{1}{2} \sum_i \dfrac{\partial E}{\partial p_i} p_i e^{-\frac{E}{kT}}$

$= \dfrac{kT}{2} \int d^f p\, d^f \xi \dfrac{1}{2} \sum_i p_i \dfrac{\partial}{\partial p_i} e^{-\frac{E}{kT}} \cdot (-kT)$

$= \dfrac{kT}{2} \dfrac{1}{h^f} \sum_i \delta_{ii} \int d\xi dp\, e^{-\frac{E}{kT}}$ $\qquad p \dfrac{\partial}{\partial p} =$

$= \dfrac{kT}{2} f$ $\qquad \dfrac{\partial}{\partial p} p = 1 + p \dfrac{\partial}{\partial p}$

19. (i) $\square_1\ \square_2\ \cdots\ \square_{G_k}$ N_k 個の分け方 $G_k^{N_k}$ 通り 粒子の区別がないから $\frac{G_k^{N_k}}{N_k!}$

各エネルギー状態が独立だから $W = \prod_k \frac{G_k^{N_k}}{N_k!}$

(ii) $S = \ln W = \sum (N_k \ln G_k - \ln N_k!) \approx \sum (N_k \ln G_k - N_k \ln N_k + N_k)$

$N = \sum N_k$

$E = \sum \epsilon_k N_k$

$\delta S = \sum (\delta N_k \ln G_k - \delta N_k \ln N_k) = 0$

$\delta N = \sum \delta N_k = 0$

$\delta E = \sum \epsilon_k \delta N_k = 0$

$\delta(S + \alpha N + \beta E) = \sum (\ln G_k - \ln N_k + \alpha + \beta \epsilon_k)\delta N_k = 0$

$\therefore \ln G_k - \ln N_k + \alpha + \beta \epsilon_k = 0$

$\therefore N_k = G_k \exp(+\alpha + \beta \epsilon_k)$

$\therefore e^{-\alpha} \sum G_k \exp(\beta \epsilon_k)$

$N = \sum N_k = \sum G_k \exp(\alpha + \beta \epsilon_k)$

$E = \sum \epsilon_k N_k = \sum \epsilon_k G_k \exp(\alpha + \beta \epsilon_k)$

$\alpha = \ln\left[\sum G_k \exp(\beta \epsilon_k)\right] - \ln N$

20. (i) $F(\alpha x_i) = \alpha^n F(x_i)$ $\rightarrow$ $\sum \frac{\partial F(\alpha x_i)}{\partial(\alpha x_i)} \frac{\partial(\alpha x_i)}{\partial \alpha} = n \alpha^{n-1} F(x_i)$ $\alpha = 1$ より o.k

(ii) (a) $T = \sum_i \frac{m_i}{2} v_i^2 = T(v_i)$ $T(\alpha v_i) = \alpha^2 T(v_i)$ T は 2次の同次式

$2T = \sum v_i \frac{\partial T}{\partial v_i} = \sum_i \dot{r}_i p_i = \frac{d}{dt}\left(\sum_i r_i p_i\right) - \sum \dot{p}_i r_i$

(b) $F(t) = \int_0^t f(t)dt$ $|F(t)| < M$ とする

$\bar{f} = \left|\frac{1}{T}\int_{-T}^{T} f(t)dt\right| = \left|\frac{F(t) - F(-T)}{T}\right| \leq \frac{|F(t)| + |F(-T)|}{T} \leq \frac{2M}{T} \to 0$

$f = \frac{\partial}{\partial t}(r_i p_i)$

$2\bar{T} = \bar{f} - \overline{\sum \dot{p}_i r_i} = \overline{\sum r_i \frac{\partial U}{\partial r_i}} = n\bar{U}$

(iii) 調和振動子

$U(r_i) = \sum_i \frac{k_i}{2}(r_i - r_j)^2$

$U(\alpha r_i) = \alpha^2 U(r_i)$

U は 2次の同次式

$2U = \sum r_i \frac{\partial U}{\partial r_i}$

$\therefore \bar{U} = \bar{T}$

万有引力

$U(r_i) = \sum_{i<j} \frac{\beta_{ij}}{|r_i - r_j|}$

$U(\alpha r_i) = \alpha^{-1} U(r_i)$

-1 次

$\sum r_i \frac{\partial U}{\partial r_i} = -U$

$\overline{2T} = -\bar{U}$

$\bar{E} = \bar{T} + \bar{U} = -\bar{T}$

$C = \frac{\partial \bar{E}}{\partial T} = -\frac{d\bar{T}}{dT} = -\frac{1}{2}Nk < 0$

$\because \bar{T} = \frac{1}{2}Nkt$

2|

(1) $F(T,V) = U - TS$

$dU = -p\,dV + T\,dS$

$-T^2\left\{\frac{\partial}{\partial T}\left(\frac{U}{T} - S\right)_V\right\} = U - T\left(\frac{\partial U}{\partial T}\right)_V + T^2\left(\frac{\partial S}{\partial T}\right)_V$ $\qquad \left(\frac{\partial U}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V$

(2) $U = \frac{\sum_r E_r e^{-\beta E_r}}{\sum_r e^{-\beta E_r}} = -\frac{\partial}{\partial \beta}(\ln Q_N)$

(3) $-\frac{\partial}{\partial \beta}(\ln Q_N) = \frac{1}{k}\frac{\partial}{\partial \beta}(F k\beta) \rightarrow F = -\frac{1}{\beta}\ln Q_N + \text{Const}$

(4) $S = \frac{1}{T}(U - F) = k\beta\left(-\frac{\partial}{\partial \beta}(\ln Q_N) + \frac{1}{\beta}\ln Q_N\right) = k \ln Q_N + \frac{kT}{Q_N}\left(\frac{\partial Q_N}{\partial T}\right)_V + \frac{\text{Const}}{T}$ $\begin{pmatrix} S \to 0 \\ T \to 0 \end{pmatrix}$

$P = -\left(\frac{\partial F}{\partial V}\right)_T = \frac{1}{\beta}\frac{1}{Q_N}\left(\frac{\partial Q_N}{\partial V}\right)_T$

$C_V = T\left(\frac{\partial S}{\partial T}\right)_V = T\frac{\partial}{\partial T}\left(k \ln Q_N + \frac{kT}{Q_N}\frac{\partial Q_N}{\partial T}\right)_V = kT\frac{\partial^2}{\partial T^2}(T \ln Q_N)$

(5) $-k\sum_r P_r \ln P_r = -k\sum_r \frac{1}{Q_N} e^{-\beta E_r}\cdot(-\beta E_r - \ln Q_N)$

$= k\beta\frac{1}{Q_N}\left(\sum_r E_r e^{-\beta E_r} + k \ln Q_N \sum_r \frac{e^{-\beta E_r}}{Q_N}\right.$

$= k \ln Q_N - k\beta\frac{1}{Q_N}\frac{\partial Q_N}{\partial \beta}$

22. $Z_i = \frac{1}{h}\int e^{-\beta H}\, dp_i\, dq_i = \frac{1}{h}\int e^{-\beta \frac{m\omega^2}{2} q_i^2} dq_i \cdot \int e^{-\beta \frac{1}{2m} p_i^2} dp_i$

$= \frac{1}{h}\sqrt{\frac{2\pi}{\beta m\omega^2}} \cdot \sqrt{\frac{2m\pi}{\beta}} = \frac{2\pi}{h\beta\omega}$ $\qquad Z = \left(\frac{2\pi}{h\beta\omega}\right)^N$

$U = -\frac{\partial}{\partial\beta} \ln Z = \frac{N}{\beta}$

$F = -\frac{1}{\beta} \ln Z = -\frac{N}{\beta} \ln\left(\frac{2\pi}{h\beta\omega}\right)$

$P = -\left(\frac{\partial F}{\partial V}\right)_T = 0$

$S = -\left(\frac{\partial F}{\partial T}\right)_V = \frac{U-F}{T} = \frac{N}{\beta T}\left(1 + \ln\frac{2\pi}{h(\beta\omega)}\right)$

$C = T\frac{\partial S}{\partial T} = Nk$

23 $z = \sum_n e^{-\beta\left(n+\frac{1}{2}\right)\hbar\omega} = 1 / 2\sinh\left(\frac{\hbar\omega}{2kT}\right)$

$Z = \left[2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right]^{-N}$

$U = -\frac{\partial}{\partial\beta} \ln Z = N\frac{\hbar\omega}{2}\coth\frac{\hbar\omega}{2kT}$

$F = -kT \ln Z = NkT \ln 2\sinh\left(\frac{\hbar\omega}{2kT}\right)$

$P = -\left(\frac{\partial F}{\partial V}\right)_T = 0$

$S = \frac{U-F}{T} = N\left[\frac{\hbar\omega}{2}\coth\left(\frac{\hbar\omega}{2kT}\right) - kT \ln\left\{2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right\}\right]$

$C_v = \left(\frac{\partial U}{\partial T}\right)_V = Nk\left(\frac{\hbar\omega}{kT}\right)^2 / \left[2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right]^2$

24. (1) $\sum_r P_r = 1$ のもとでの極値 (Lagrange 未定係数法)

$$S = -k \sum_r P_r \ln P_r + \alpha \left(1 - \sum_r P_r\right)$$

S が P_r^* で極値をとる

$$\frac{\partial S}{\partial P_r^*} = -\sum_r \left(k \ln P_r^* + k + \alpha\right) = 0 \qquad \left(\frac{\partial^2 S}{\partial P_r^{*2}} = -\sum_r \frac{k}{P_r} < 0 \quad \text{極大}\right)$$

$$P_r^* = e^{-\left(1+\frac{\alpha}{k}\right)}$$

$$1 = \sum_r P_r^* = \sum_r e^{-\left(1+\frac{\alpha}{k}\right)} = \left(\sum_r 1\right) e^{-\left(1+\frac{\alpha}{k}\right)} \qquad \left(\sum_r 1 = \Omega\right)$$

$$P_r^* = e^{-\left(1+\frac{\alpha}{k}\right)} = \frac{1}{\Omega}$$

(2) $\sum_r P_r = 1$, $\sum_r P_r \varepsilon_r = E$

$$S = -k \sum_r P_r \ln P_r + \alpha \left(1 - \sum_r P_r\right) + \beta \left(E - \sum_r P_r \varepsilon_r\right)$$

同様にして $P_r^* = \exp\left(-1 - \frac{\alpha}{k} - \frac{\beta}{k} \varepsilon_r\right)$

$$1 = \sum_r P_r^* = \exp\left(-1 - \frac{\alpha}{k}\right) \sum_r \exp\left(-\frac{\beta}{k} \varepsilon_r\right)$$

$$P_r^* = \frac{1}{Z(\beta)} \exp\left(\frac{\beta}{k} \varepsilon_r\right)$$

(3) $\sum_r P_r = 1$, $\sum_r P_r \varepsilon_r = E$, $\sum_r P_r n_r = N$

$$S = -k \sum_r P_r \ln P_r + \alpha \left(1 - \sum_r P_r\right) + \beta \left(E - \sum_r P_r \varepsilon_r\right) + \gamma \left(N - \sum_r P_r n_r\right)$$

$$\rightarrow P_r^* = \exp\left(1 - \frac{\alpha}{k} - \frac{\beta}{k} \varepsilon_r - \frac{\gamma}{k} n_r\right)$$

25

(1) $S = -k \sum_{r,s} P_r^{(1)} P_s^{(2)} \left(\ln P_r^{(1)} + \ln P_s^{(2)} \right)$

$= -k \sum_r P_r^{(1)} \ln P_r^{(1)} \sum_s P_s^{(2)} - k \sum_s P_s^{(2)} \ln P_s^{(2)} \sum_r P_r^{(1)}$

$= -k \sum_r P_r^{(1)} \ln P_r^{(1)} - k \sum_s P_s^{(2)} \ln P_s^{(2)} = S_1 + S_2$

(2) $S_1 + S_2 - S = -k \sum_r P_r^{(1)} \ln P_r^{(1)} - k \sum_s P_s^{(2)} \ln P_s^{(2)} + k \sum_{r,s} P_{rs} \ln P_{rs}$

$= -k \sum_r \left(\sum_s P_{rs} \right) \ln P_r^{(1)} - k \sum_s \left(\sum_r P_{rs} \right) \ln P_s^{(2)} + k \sum_{r,s} P_{rs} \ln P_{rs}$

$= -k \sum_{r,s} \left(P_{rs} \ln P_r^{(1)} + P_{rs} \ln P_s^{(2)} - P_{rs} \ln P_{rs} \right)$

$= -k \sum_{r,s} P_{rs} \ln \left\{ \frac{P_r^{(1)} P_s^{(2)}}{P_{rs}} \right\} \geq -k \sum_{r,s} \left(\frac{P_r^{(1)} P_s^{(2)}}{P_{rs}} - 1 \right)$

$= -k \sum_{r,s} \left(P_r^{(1)} P_s^{(2)} - P_{rs} \right) = 0$

$\therefore S \leq S_1 + S_2$

$P_r^{(1)} P_s^{(2)} - \Sigma$

26.

1原子の内部エネルギー ε_l の状態にある確率 $P = \dfrac{g_l e^{-\frac{\varepsilon_l}{kT}}}{\sum_l g_l e^{-\frac{\varepsilon_l}{kT}}}$

g_l: 縮重度, l 量子数

$Z = 1 \cdot e^{-\beta E_0} + 3e^{-\beta(E_0 + 1500\,k_B)}$

$\Xi = -\frac{\partial}{\partial \beta} \ln Z = 750\,k_B$ ✓ $\frac{1500\,k_B}{2}$ (平均)

$\dfrac{4500\,k_B\, e^{-\frac{1500}{T}}}{1 - e^{-\frac{1500}{T}}} = 750\,k_B \quad \rightarrow T = 1365\ \mathrm{K}$

$\dfrac{e^{-\frac{\varepsilon_0}{kT}}}{e^{-\frac{\varepsilon_0}{kT}}} = \dfrac{}{1 + }$

H atom 基底状態 $-13.6\ eV$

第一励起状態 $-3.4\ eV$

$10.2\ eV = k_B T_1 \quad \rightarrow T_1 = 118200\ \mathrm{K}$

$P_0 \propto e^{-\beta E_0}$

$P_1 \propto 3e^{-\beta E_1}$

$\begin{cases} e^{-\beta E_0} = 3e^{-\beta E_1} \\ E_1 - E_0 = 1500\,k_B \end{cases}$

$-\beta E_0 = -\beta E_1 + \ln 3$

$\beta(E_1 - E_0) = \ln 3$

$T = \dfrac{1500}{\ln 3} = 1365$

27.

(1) $H = \sum_{i=1}^{3N} \frac{p_i^2}{2m}$ $\quad Z = \frac{1}{h^{3N} N!} \int e^{-\beta H} d\Gamma$, $d\Gamma = \prod_{i=1}^{3N} dq_i dp_i$

$$= \frac{V^N}{h^{3N} N!} (2\pi m k T)^{\frac{3N}{2}}$$

$$F = -kT \ln Z \sim -NkT \left\{ 1 + \frac{3}{2} \ln T + \ln \frac{V}{N} + \ln \frac{(2\pi mk)^{\frac{3}{2}}}{h^3} \right\}$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V = -Nk \left\{ \frac{5}{2} + \frac{3}{2} \ln T + \ln \frac{V}{N} + \ln \frac{(2\pi mk)^{\frac{3}{2}}}{h^3} \right\}$$

$$N! = \left(\frac{N}{e}\right)^N$$

(2) $T_{Ar} \cdot m_{Ar} = T_{He} \cdot m_{He}$

$300\,K \cdot 40 = T_{He} \cdot 4$

$T_{He} = 3000\,K$

28. $Z = (e^0 + e^{-\varepsilon/kT})^N = (1 + e^{-\frac{\varepsilon}{kT}})^N$

(1) $F = -NkT \ln(1 + e^{-\frac{\varepsilon}{kT}})$

$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V} = -kT \ln(1 + e^{-\frac{\varepsilon}{kT}})$

(2) $U = -T^2 \frac{\partial}{\partial T}\left(\frac{F}{T}\right) =$

$S = -\left(\frac{\partial F}{\partial T}\right)_V =$

(3) $C = \left(\frac{\partial U}{\partial T}\right)_V =$

$-N \frac{\partial}{\partial \beta} \ln(1 + e^{-\frac{\beta\varepsilon}{kT}})$

$-N \frac{-\varepsilon}{1 + e^{-\beta\varepsilon}}$

$\frac{1}{1 + e^{-\frac{\varepsilon}{kT}}}$

$1 - \frac{\varepsilon}{kT}$

$\frac{N\varepsilon}{2 - \frac{\varepsilon}{kT}}$

$\ln 2$

$e^{\frac{\varepsilon}{kT}}$

$T \to 0$

$e^{x \to \infty}$

29.

$$Z = \frac{V^N}{h^{3N} N!} (2\pi m k T)^{\frac{3N}{2}} \quad , \quad E = -\frac{\partial}{\partial \beta}(\ln Z) = \frac{3}{2} N k T$$

$$PV = NkT \rightarrow E = \frac{3}{2} PV$$

$$\frac{3}{2} P \cdot V + 10^4 [J] = 1.19 \times 10^3 + 10^4 [J] = 1.119 \times 10^4 [J]$$

$$T = 2.82 \times 10^3 [K]$$

$$\left(\frac{3}{2} PV\right) + 10^4 = \left(\frac{3}{2} \frac{PV}{\cancel{0}}\right) \frac{T'}{T}$$

$$\frac{T'}{T} = 1 + \frac{2}{3PV} \cdot 10^4$$

$$\frac{2}{3 \cdot (0.1)^2 \pi \cdot 10^{+5}} \cdot 10^4$$

$$\frac{2}{3 \times 0.01 \times 3 \times 10} : \frac{2}{0.1 \times 9}$$

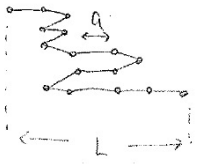
$V = \pi 0.1^2 \cdot 1$

$= 0.03$



9.8

30.



$L = 2Ma \quad (M = 0, 1, \cdots, N)$

$\ln x! = x \ln \frac{x}{e}$

$++--+-+$

$\oplus\ n_+$ コ $\ominus\ n_-$ コ

$\begin{cases} n_+ + n_- = 2N \\ n_+ - n_- = 2M \end{cases}$

(1) ~~2N コ a で 2M コ 分配~~

$$\frac{(2N)!}{(N-M)!\,(N+M)!}$$

$$W(L) = \frac{(2N)!}{(N-\frac{L}{2a})!\,(N+\frac{L}{2a})!}$$

$$S = k \ln W(L) \simeq k \left\{ 2N \ln \frac{2N}{e} - (N-\tfrac{L}{2a}) \ln \frac{N-\frac{L}{2a}}{e} - (N+\tfrac{L}{2a}) \ln \frac{N+\frac{L}{2a}}{e} \right\}$$

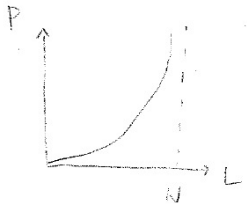
$$= k \left\{ 2N \ln 2N - (N-\tfrac{L}{2a}) \ln (N-\tfrac{L}{2a}) - (N+\tfrac{L}{2a}) \ln (N+\tfrac{L}{2a}) \right\}$$

(2) $F = -kT \ln W(L)$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V = k \ln W(L)$$

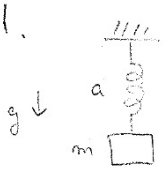
(3) $$P = \left(\frac{\partial F}{\partial L}\right)_{T,V} = -T\left(\frac{\partial S}{\partial L}\right)_T = kT \left\{ -\frac{1}{2a} \ln (N-\tfrac{L}{2a}) + \frac{1}{2a} \ln (N+\tfrac{L}{2a}) \right\}$$

$$= \frac{kT}{2a} \ln \frac{2Na+L}{2Na-L}$$



P = const で T を上げると 広がる

31.



$$\frac{1}{2}ax^2 - mgx = \frac{1}{2}a\left(x - \frac{mg}{a}\right)^2 - \frac{m^2g^2}{2a}$$

(1) $$\bar{x} = \frac{\int_{-\infty}^{\infty} x\, e^{-\frac{1}{kT}\left(\frac{1}{2}ax^2 - mgx\right)} dx}{\int_{-\infty}^{\infty} e^{-\frac{1}{kT}\left(\frac{1}{2}ax^2 - mgx\right)} dx} = \frac{mg}{a}$$

(2) $$(x - \bar{x})^2 = \frac{\int_{-\infty}^{\infty} \left(x - \frac{mg}{a}\right)^2 e^{-\frac{1}{kT}\left(\frac{1}{2}a\left(x - \frac{mg}{a}\right)^2 - \frac{m^2g^2}{2a}\right)} dx}{\int_{-\infty}^{\infty} e^{-\frac{1}{kT}\left(\frac{1}{2}a\left(x - \frac{mg}{a}\right)^2 - \frac{m^2g^2}{2a}\right)} dx}$$

$$= \int_{-\infty}^{\infty} x^2 e^{-\frac{a}{2kT}x^2} dx \Big/ \int_{-\infty}^{\infty} e^{-\frac{a}{2kT}x^2} dx = \frac{kT}{a}$$

$$\int e^{-ax^2} = \left(\frac{\pi}{a}\right)^{\frac{1}{2}}$$

$$\int -x^2 e^{-ax^2} = -\frac{1}{2\pi}\left(\frac{\pi}{a}\right)^{\frac{3}{2}} = -\frac{1}{2a}\sqrt{\frac{\pi}{a}}$$

(3) $$\sqrt{(x - \bar{x})^2} = \sqrt{\frac{kT}{a}} \ll \frac{Mg}{a}$$

32.

$$E = -MH = -\mu H \sum_{j=1}^{N} m_j \qquad (m_j = -J, \cdots, J)$$

$$Z = \sum_{m_1=-J}^{J} \cdots \sum_{m_N=-J}^{J} e^{\frac{\mu H}{kT}\sum_{j=1}^{N} m_j} =$$

---

(1) $$Z = \left(\sum_{j=-J}^{J} e^{\beta J \mu H} \right)^N = e^{-\beta J \mu H} \frac{1 - e^{\beta(2J+1)\mu H}}{1 - e^{\beta \mu H}} \qquad \ln Z = N \ln \frac{\sinh\left(J+\frac{1}{2}\right)\beta\mu H}{\sinh\left(\frac{1}{2}\beta\mu H\right)}$$

$$\bar{E} = -\frac{\partial}{\partial \beta} \ln Z = -N\mu H \left[\left(J+\tfrac{1}{2}\right) \coth\left(J+\tfrac{1}{2}\right)\beta\mu H - \tfrac{1}{2}\coth\left(\tfrac{1}{2}\beta\mu H\right) \right]$$

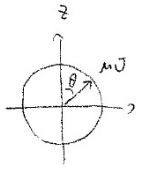
$$\bar{E} = -\bar{M}H \rightarrow \bar{M} = N\mu \left[\left(J+\tfrac{1}{2}\right) \coth\left(J+\tfrac{1}{2}\right)\beta\mu H - \tfrac{1}{2}\coth\left(\tfrac{1}{2}\beta\mu H\right) \right]$$

(2) $\beta\mu H \ll 1 \qquad \coth x = \frac{1}{x} + \frac{x}{3} + O(x^3)$

$$\bar{M} \sim N\mu \left[\left(J+\tfrac{1}{2}\right) \left\{ \frac{1}{\left(J+\frac{1}{2}\right)\beta\mu H} + \frac{\left(J+\frac{1}{2}\right)\beta\mu H}{3} \right\} - \frac{1}{2}\left\{ \frac{1}{\frac{1}{2}\beta\mu H} + \frac{\frac{1}{2}\beta\mu H}{3} \right\} \right]$$

$$= \frac{N\mu^2\beta H}{3} J(J+1) \qquad \chi = \left.\frac{\partial \bar{M}}{\partial H}\right|_{H=0} = \frac{1}{3} J(J+1) \frac{N\mu^2}{kT}$$

(3) $$Z = \left(\int e^{\beta J \mu H \cos\theta} \frac{d\Omega}{4\pi} \right)^N = \left(\frac{\sinh \beta J \mu H}{\beta J \mu H} \right)^N$$



$$\bar{E} = -N\left[J\mu H \coth \beta J \mu H - \frac{1}{\beta} \right]$$

$$\bar{M} = N\left[J\mu \coth \beta J \mu H - \frac{1}{\beta H} \right]$$

$\beta\mu H \ll 1$

$$\bar{M} = \frac{N\mu^2 J^2 \beta}{3} H \qquad \chi = \frac{N\mu^2 J^2}{3kT}$$

33

$$Z(\beta) = \sum e^{-\beta E_\ell} = \int e^{-\beta E\,\Omega(E)}\,dE$$

$$\overline{E^n} = \int (E\,\Omega(E))^n e^{-\beta E\,\Omega(E)}\,dE \,/ Z = (-1)^n \frac{Z^{(n)}}{Z} \qquad (n) \equiv \frac{\partial^n}{\partial \beta^n}$$

$$\frac{\partial \bar{E}}{\partial \beta} = -\frac{Z''}{Z} + \frac{Z'^2}{Z^2} = -\bar{E}^2 + \bar{E}^2 = -\overline{(E-\bar{E})^2}$$

$$\frac{\partial^2 \bar{E}}{\partial \beta^2} = -\frac{Z'''}{Z} + 3\frac{Z'Z''}{Z^2} - 2\frac{Z'^3}{Z^3} = \overline{E^3} - 3\overline{E^2}\,\bar{E} + 2\bar{E}^3 = \overline{(E-\bar{E})^3}$$

$$\frac{\partial^3 \bar{E}}{\partial \beta^3} = -\frac{Z''''}{Z} + 4\frac{Z'''Z'}{Z^2} - 12\frac{Z''Z'^2}{Z^3} + 3\frac{Z''^2}{Z^2} + 6\frac{Z'^4}{Z^4}$$

$$= -\overline{E^4} + 4\overline{E^3}\,\bar{E} - 12\overline{E^2}\,\bar{E}^2 + 3\overline{E^2} + 6\bar{E}^4$$

$$= -\overline{(E-\bar{E})^4} + 3\left(\frac{\partial \bar{E}}{\partial \beta}\right)^2$$

$$\frac{\partial \bar{E}}{\partial \beta} = -kT^2\frac{\partial \bar{E}}{\partial T} = -kT^2 C_v \qquad \left(\frac{\partial}{\partial \beta} = -kT^2\frac{\partial}{\partial T}\right)$$

$$\frac{\partial^2 \bar{E}}{\partial \beta^2} = k^2T^2\frac{\partial}{\partial T}\left(T^2 C_v\right) = k^2\left\{T^4\frac{\partial C_v}{\partial T} + 2T^3 C_v\right\}$$

$$\frac{\partial^3 \bar{E}}{\partial \beta^3} = -k^3T^2\frac{\partial}{\partial T}\left(T^4\frac{\partial C_v}{\partial T} + 2T^3 C_v\right) = -k^3\left\{T^6\frac{\partial^2 C_v}{\partial T^2} + 6T^5\frac{\partial C_v}{\partial T} + 6T^4 C_v\right\}$$

(11) $$\overline{(E-\bar{E})^4} = \frac{\partial^3 \bar{E}}{\partial \beta^3} + 3\left(\frac{\partial \bar{E}}{\partial \beta}\right)^2 = k^3\left\{T^6\frac{\partial^2 C_v}{\partial T^2} + 6T^5\frac{\partial C_v}{\partial T} + 6T^4 C_v\right\} + 3k^2T^4C_v^2$$

(=) $$\bar{E} = \frac{3}{2}NkT, \quad C_v = \frac{3}{2}Nk$$

$$kT^2C_v = k\left(\frac{\bar{E}}{C_v}\right)^2 C_v = \bar{E}^2\frac{2}{3N}$$

$$k^2\left\{T^4\frac{\partial C_v}{\partial T} + 2T^3 C_v\right\} = 2k^2\left(\frac{\bar{E}}{C_v}\right)^3 C_v = \bar{E}^3\frac{8}{9N^2}$$

($\frac{\partial C_v}{\partial T} = 0$)

34.

$$\Xi = \sum_{N,\tau} e^{\beta(\mu N - \epsilon N)}$$

$$\langle N \rangle = \frac{1}{\beta} \frac{\partial \Xi}{\partial \mu} / \Xi$$

1. $$\langle N^2 \rangle - \langle N \rangle^2 = \frac{1}{\beta^2}\left[\frac{\partial^2 \Xi}{\partial \mu^2} / \Xi - \left(\frac{\partial \Xi}{\partial \mu} / \Xi \right)^2 \right] = \frac{1}{\beta^2} \frac{\partial}{\partial \mu}\left(\frac{\partial \Xi}{\partial \mu} / \Xi \right) = \frac{1}{\beta} \frac{\partial}{\partial \mu} \langle N \rangle$$

0. $$\Xi = \sum \lambda^N Z = \sum \lambda^N \frac{Z_1^N}{N!} = e^{\lambda Z_1} \quad (\lambda = e^{\beta\mu})$$

$$\langle N \rangle = \frac{\partial \Xi}{\partial \lambda} / \Xi = \frac{\partial}{\partial \lambda} \ln \Xi = \lambda Z_1$$

$$\langle N^2 \rangle - \langle N \rangle^2 = \frac{1}{\beta} \frac{\partial}{\partial \mu} \langle N \rangle = \lambda Z_1$$

11. $$\Xi_\tau = \sum_n e^{\beta(\mu - \epsilon_\tau) n} = \frac{1}{1 \pm e^{\beta(\mu - \epsilon_\tau)}}$$

$$\langle n_\tau \rangle = \frac{1}{\beta} \frac{\partial}{\partial \mu} \Xi_\tau / \Xi_\tau = \frac{1}{e^{\beta(\mu - \epsilon_\tau)} \pm 1}$$

$$\langle n_\tau^2 \rangle - \langle n_\tau \rangle^2 = \frac{1}{\beta} \frac{\partial}{\partial \mu} \left(\frac{1}{e^{\beta(\mu - \epsilon_\tau)} \pm 1} \right)$$

$$= \frac{1}{\beta} \langle n_\tau \rangle^2 \left(-\beta e^{\beta(\epsilon_\tau - \mu)} \right)$$

$$=$$

$$e^{\beta(\epsilon - \mu)} \mp 1 = \frac{1}{\langle n \rangle}$$

$$e^{\beta(\epsilon - \mu)} = \frac{1}{\langle n \rangle} \pm 1$$

$$- \langle n \rangle^2 \left(\frac{1}{\langle n \rangle} \pm 1 \right)$$

$$- \langle n \rangle \left(1 \pm \langle n \rangle \right)$$

35

[1] $z_n = \frac{d^n}{dt^n} g(t)\Big|_{t=0} = \frac{d^n}{dt^n}\left(\log f(t)\right)\Big|_{t=0}$ ※)

$\left(f(t) = \langle e^{tx}\rangle,\ f^{(n)}(t)|_{t=0} = \langle x^n\rangle\right)$

$z_1 = \frac{f'(t)}{f(t)}\Big|_{t=0} = \langle x\rangle$, $z_2 = \frac{f''}{f} - \frac{f'^2}{f^2} = \langle x^2\rangle - \langle x\rangle^2$

$z_3 = \frac{f'''}{f} - 3\frac{f''f'}{f^2} + 2\frac{f'^3}{f^3} = \langle x^3\rangle - 3\langle x^2\rangle\langle x\rangle + 2\langle x\rangle^3$

[2] $f(t) = \langle e^{tx}\rangle = \int_{-\infty}^{\infty} e^{tx}\cdot\frac{1}{\sqrt{2\pi v}} e^{-\frac{(x-m)^2}{2v}} dx = \underbrace{e^{mt+\frac{v}{2}t^2}}\cdot\underbrace{\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi v}} e^{-\frac{\{x-(m+vt)\}^2}{2v}} dx}_{1}$

$g(t) = \log f(t) = \underset{z_1}{mt} + \underset{z_2}{\frac{1}{2}vt^2}$]

$z_{n\geq 3} = 0$

[3] $f(t) = \sum_{n=0}^{\infty} e^{tn}\cdot\frac{a^n e^{-a}}{n!} = e^{-a}\sum_n \frac{1}{n!} e^{(\ln a + t)n} = e^{-a} e^{e^{(\ln a + t)}}$

$= e^{-a} e^{ae^t} = e^{a(e^t-1)}$

$g(t) = a(e^t-1) = \sum_{n=1}^{\infty}\frac{t^n}{n!} a$ $z_n = a$ const with n

$(e^t\cdot a)^x$

$e^{-a} e^{e^t a}$

$e^t = \sum\frac{1}{n!} t^n$

36

$$Z_1 = \frac{1}{h^3}\int e^{-\beta \frac{\vec{p}^2}{2m}} d^3p\, d^3q = \frac{V}{h^3}\sqrt{2\pi m kT}^3$$

$$F = -kT \ln \frac{V^N (2\pi m kT)^{\frac{3}{2}N}}{h^{3N} N!} \simeq -NkT\left\{\ln V + \ln(2\pi m kT)^{\frac{3}{2}} - \ln h^3 - \ln N + 1\right\}$$

$$\mu = \frac{\partial F}{\partial N} = -kT\left\{\ln V + \ln(2\pi m kT)^{\frac{3}{2}} - \ln h^3 - \ln N\right\}$$

$$= kT \ln\left[\left(\frac{N}{V}\right)\cdot\left(\frac{h^2}{2\pi m kT}\right)^{\frac{3}{2}}\right]$$

$$dF = -S\,dT + X\,dL$$

37 $\quad X = AT = \left(\frac{\partial F}{\partial L}\right)_T \quad \rightarrow \quad F = ATL + f(T)$

$$S = -\left(\frac{\partial F}{\partial T}\right)_L = -AL - \left(\frac{\partial f}{\partial T}\right)_L = -AL - f'(T)$$

$$U = F + TS = f(T) - Tf'(T)$$

$$N! = \left(\frac{N}{e}\right)^N$$

[38]

$$g(\omega) = \begin{cases} \dfrac{9N}{\omega_D^3}\omega^2 & (0<\omega\le\omega_D) \\ 0 & (\omega_D<\omega) \end{cases}$$

$$Z = \prod_{\omega}\left(\frac{1}{1-e^{-\beta\hbar\omega}}\right)^{g(\omega)\Delta\omega} \quad\to\quad \ln Z = -\int g(\omega)\ln\left(1-e^{-\beta\hbar\omega}\right)d\omega$$

$$E = -\frac{\partial}{\partial\beta}(\ln Z) = \int g(\omega)\frac{\hbar\omega e^{-\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}}d\omega = \frac{9N}{\omega_D^3}\int_0^{\omega_D}\frac{\hbar\omega^3}{e^{\beta\hbar\omega}-1}d\omega$$

$\beta\hbar\omega = x$, $d\omega = \frac{1}{\beta\hbar}dx$

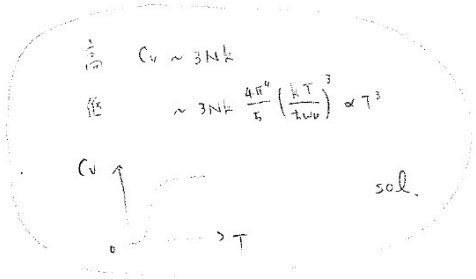
$$= \frac{9N}{\beta^4\hbar^3\omega_D^3}\cdot\int_0^{\beta\hbar\omega_D}\frac{t^3}{e^t-1}dt$$

$$= 3NkT\cdot D(\beta\hbar\omega_D)$$

$$D(x) = \frac{3}{x^3}\int_0^x dt\,\frac{t^3}{e^t-1}$$

デバイ関数

$$C_V = \frac{\partial E}{\partial T} = 3Nk\,D\left(\frac{\hbar\omega_D}{kT}\right) - \frac{3N\hbar\omega_D}{T}D'\left(\frac{\hbar\omega_D}{kT}\right)$$



$$\frac{x^3}{e^x-1} = \left(\frac{x^3}{3}D(x)\right)'$$

$$= x^3 D(x) + \frac{x^3}{3}D'(x)$$

$$D'(x) = -\frac{3}{x}D(x) + \frac{3}{e^x-1}$$

$$C_V = 3Nk\left(D(x) - xD'(x)\right)$$

$$= 3Nk\left(4D(x) - \frac{3x}{e^x-1}\right)$$

$$D(x) \to \frac{3}{3x^2}\cdot\frac{x^3}{e^x-1} \to 1 \quad (x\to 0)$$

$$D(x) \to \frac{3}{x^3}M \quad (x\to\infty)$$

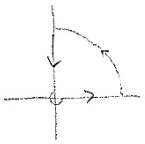
$$M = \int_0^\infty dt\,\frac{t^3}{e^t-1} =$$

"

$$t + \frac{1}{2}t^2 + \frac{1}{6}t^3 + \frac{1}{24}t^4 + \frac{1}{120}t^5$$

$T\to\frac{1}{60}$ $(x\to 0)$ $\quad C_V = 3Nk\cdot(4\cdot 1 - 3) = 3Nk$

$T\to$ 低温 $(x\to\infty)$ $\quad C_V = 3Nk\cdot\frac{12}{x^3}M$



[39]

classical $\Xi = \sum_N \frac{1}{N! h^{3N}} \int e^{-\beta(H_N - N\mu)} d\Gamma$

quantum $\Xi = \sum_N \sum_s e^{-\beta(E_{N,s} - N\mu)}$ } def

(1) $\Omega = -PV = -\frac{1}{\beta} \ln \Xi$ $\quad d\Omega = -S\,dT - P\,dV - N\,d\mu$

$\bar{N} = -\left(\frac{\partial \Omega}{\partial \mu}\right)_{V,T} = \frac{1}{\beta} \frac{\partial}{\partial \mu} \ln \Xi$ $\quad \bar{P} = -\left(\frac{\partial \Omega}{\partial V}\right)_{T,\mu} = \frac{1}{\beta} \frac{\partial}{\partial V} \ln \Xi$

$\Xi = \sum_N \sum_r e^{\beta(\mu N - E_r)} \rightarrow \bar{E} = \frac{1}{\Xi} \sum_N \sum_r E_r e^{\beta(\mu N - E_r)}$

$= -\frac{1}{\Xi} \frac{\partial}{\partial \beta} \Xi + \frac{1}{\Xi} \sum_N \sum_r \mu N e^{\beta(\mu N - E_r)}$

$\bar{N}\mu$

$= -\frac{1}{\Xi} \frac{\partial}{\partial \beta} \Xi + \frac{\mu}{\beta} \frac{\partial}{\partial \mu} \ln \Xi$

(2) $\Xi = \sum_N \frac{1}{N! h^{3N}} \int e^{\beta\left(\mu N - \frac{\sum p^2}{2m}\right)} d^{3N}p\, d^{3N}x$

$= \sum_N \frac{V^N}{N! h^{3N}} e^{\beta\mu N} \left(\frac{2\pi m}{\beta}\right)^{\frac{3N}{2}}$ $\quad \lambda_T^2 = \frac{\beta h^2}{2\pi m}$

$= \sum_N \frac{V^N}{N!} e^{\beta\mu N} \lambda_T^{-3N} = \sum_N \frac{1}{N!} \left(\frac{V e^{\beta\mu}}{\lambda_T^3}\right)^N = \exp\left(\frac{V e^{\beta\mu}}{\lambda_T^3}\right)$

$\bar{N} = kT \frac{\partial}{\partial \mu} \ln \Xi = \frac{V e^{-\beta\mu}}{\lambda_T^3}$

$\bar{P} = kT \frac{\partial}{\partial V} \ln \Xi = \frac{\bar{N} kT}{V}$

$\bar{E} = -\frac{1}{\Xi} \frac{\partial}{\partial \beta} \Xi + \mu \bar{N} = \ldots = \frac{3}{2} \bar{N} kT$

[41] (その1)

電子スピン $\frac{1}{2}\hbar$, $-\frac{1}{2}\hbar$: s $\quad \mu = \frac{es}{mc}$ Fermion

ボーア磁子 $\mu_B = \frac{e\hbar}{2mc}$

$$Z_1 = e^{-\beta\mu_B H} + e^{\beta\mu_B H} = 2\cosh(\beta\mu_B H) \qquad Z = Z_1^N$$

1個あたり $\langle \mu \rangle = \frac{1}{Z_1}(-\mu_B e^{-\beta\mu_B H} + \mu_B e^{\beta\mu_B H})$

$$= \frac{\partial}{\partial(\beta H)} \ln Z_1 = \frac{\partial}{\partial(\beta H)} \ln\left[2\cosh(\beta\mu_B H)\right] = \mu_B \tanh(\beta\mu_B H)$$

$$M = N\langle \mu \rangle = N\mu_B \tanh(\beta\mu_B H)$$

[42]

$$\Xi = \sum_N \sum_r e^{\beta(\mu N - E_r)} \qquad N = \sum n_i \ , \ E = \sum n_i \varepsilon_i$$

$$= \sum_N \left[\sum_r e^{\beta(\mu \sum n_i - \sum n_i \varepsilon_i)} \right]$$

$$= \sum_N \left[\sum_r e^{\sum \beta n_i(\mu - \varepsilon_i)} \right] = \sum_{n_1} \sum_{n_2} \cdots \sum_{n_i} \cdots e^{\sum \beta n_i(\mu - \varepsilon_i)}$$

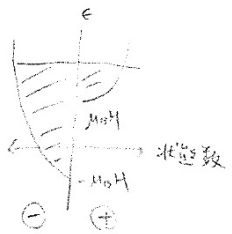
Bose $n_i = 0, 1, 2, \cdots, \infty$

[41] (3.2)

spin ± に対して $\epsilon = \frac{p^2}{2m} \pm \mu_B H$

状態密度 $D(\epsilon) = 2 \cdot 2\pi \frac{V}{h^3} (2m)^{3/2} \epsilon^{1/2}$

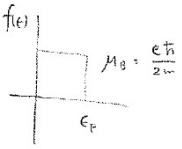
$T = 0°K$;



$$N_- = \int_0^{\epsilon_F + \mu_B H} \frac{1}{2} D(\epsilon)\, d\epsilon = \frac{4\pi V}{3h^3} \{2m(\epsilon_F + \mu_B H)\}^{3/2}$$

$$N_+ = \int_0^{\epsilon_F - \mu_B H} \frac{1}{2} D(\epsilon)\, d\epsilon = \frac{4\pi V}{3h^3} \{2m(\epsilon_F - \mu_B H)\}^{3/2}$$

$\epsilon_F \gg \mu_B H$



$$M = -2\mu_B S = -\mu_B (N_+ - N_-) = -\mu_B \frac{4\pi V^3}{h^3 \epsilon_F} \cdot \mu_B H (2m\epsilon_F)^{3/2} = \frac{3}{2} \frac{\mu_B^2 N}{\epsilon_f} H$$

$$\chi = \frac{3}{2} \frac{\mu_B^2 N}{\epsilon_F}$$

$$\left(N = \int_0^{\epsilon_F} D(\epsilon)\, d\epsilon = \frac{8\pi V}{3h^3} (2m\epsilon_F)^{3/2} \quad (= N_+ + N_-) \right)$$

古典論 $N_+ : N_- = e^{-\frac{\mu_B H}{k_B T}} : e^{\frac{\mu_B H}{k_B T}} \rightarrow \chi = \frac{M}{H} = \frac{\mu_B^2 N}{k_B T}$

$T > 0$ $\frac{1}{2} D(\epsilon) = g(\epsilon)$ とおき

$$M = \mu_B \left\{ \int \frac{g(\epsilon)\, d\epsilon}{e^{\beta(\epsilon - \mu_B H - \mu)} + 1} - \int \frac{g(\epsilon)\, d\epsilon}{e^{\beta(\epsilon + \mu_B H - \mu)} + 1} \right\}$$

$$= \mu_B \int_0^\infty d\epsilon \{ g(\epsilon + \mu_B H) - g(\epsilon - \mu_B H) \} f(\epsilon)$$

$$= 2\mu_B^2 H \int_0^\infty g'(\epsilon) f(\epsilon)\, d\epsilon \qquad (g(0) = 0)$$

$$\chi = 2\mu_B^2 \left\{ \int_0^\mu g'(\epsilon)\, d\epsilon + g''(\mu) \frac{\pi^2}{6} (kT)^2 \cdots \right\}$$

$$= 2\mu_B^2 \left\{ g(\mu) + g''(\mu) \frac{\pi^2}{6} (kT)^2 \cdots \right\}$$

$$= 2\mu_B^2 g(\epsilon_F) \left\{ 1 + O\left(\left(\frac{k_B T}{\epsilon_F} \right)^2 \right) \right\}$$

$g(\epsilon \pm \mu H) = g(\epsilon) \pm \mu H g'(\epsilon)$

$$f(\epsilon) = \frac{1}{e^{\frac{\epsilon - \mu}{kT}} + 1}$$

∴ spin による単純常磁性率は
金属の伝導電子について温度には
ほとんど依存しない

[42]

(1) $F' = U' - TS'$

$= -M\varepsilon - \frac{1}{\beta} \ln \left(\frac{P}{M}\right)$

$= -M\varepsilon - \frac{1}{\beta}\left[P \ln P - M \ln M - (P-M)\ln(P-M)\right]$

$\mu' = \frac{\partial F'}{\partial M} = -\varepsilon + \frac{1}{\beta} \ln \frac{M}{P-M}$

(2) $\mu = \frac{1}{\beta} \ln \left[\frac{N}{V}\left(\frac{h^2\beta}{2\pi m}\right)^{3/2}\right]$

$\mu' = \mu \quad \rightarrow \quad \frac{1}{\beta} \ln \left[\frac{Q-\theta P}{V}\left(\frac{h^2\beta}{2\pi m}\right)^{3/2} \cdot \frac{1-\theta}{\theta}\right] = -\varepsilon \qquad \left(\theta = \frac{M}{P}\right)$

$(Q-\theta P)(1-\theta) = \theta\alpha \quad ; \quad \alpha = V\left(\frac{2\pi m}{h^2\beta}\right)^{3/2} e^{-\beta\varepsilon}$

低温 $\alpha \ll 1, \Rightarrow \theta = 1 \quad (P \ll Q)$ or $\frac{Q}{P} \quad (P > Q)$

高温 $\alpha \gg 1 \Rightarrow \theta = \frac{Q}{P+Q+\alpha} \simeq \frac{Q}{Q+\alpha}$

$Z = \sum e^{-\beta E}$

$\Xi = \sum e^{\mu M} \, p^{C_M} \, e^{\beta M \varepsilon}$

$\langle M \rangle = \frac{1}{\beta} \frac{\partial}{\partial \mu} \ln \Xi$

$\theta = \frac{\langle M \rangle}{P} = \frac{1}{e^{-\beta(\varepsilon-\mu)} + 1}$

[43]

$\varepsilon_s = -\varepsilon + \hbar\omega\left(s+\frac{1}{2}\right)$

(1) $\Xi_s = \sum_{N=0}^{\infty} \lambda^N Z_s^N$ $\qquad Z_s = \sum e^{-\beta\varepsilon_s} = e^{\beta\varepsilon}\cdot\frac{e^{-\frac{1}{2}\beta\hbar\omega}}{1-e^{\beta\hbar\omega}} = \frac{e^{\beta\varepsilon}}{2}\operatorname{cosech} e^{\frac{1}{2}\beta\hbar\omega}$?

$= \frac{1}{1-\lambda_s Z_s}$ $\qquad \lambda_s = e^{\beta\mu_s}$

(2) $N_s = \lambda_s \frac{\partial}{\partial\lambda_s}\log \Xi_s = \frac{\lambda_s Z_s}{1-\lambda_s Z_s}$ $\qquad \therefore \lambda_s = \frac{1}{Z_s}\cdot\frac{N_s}{N_s+1} \simeq \frac{1}{Z_s}$

(3) [36] $\quad N_g = \lambda_g Z_g \quad ; Z_g = V\left(\frac{m}{2\pi\beta\hbar^2}\right)^{3/2}$ $\qquad \therefore \lambda_g = \frac{N_g}{Z_g}$

(4) $\lambda_s = \lambda_g \qquad N_g = \frac{Z_g}{Z_s}$

$$\frac{N_g}{V} = \left(\frac{m}{2\pi\beta\hbar^2}\right)^{3/2} e^{-\beta\varepsilon}\, 2\sinh\tfrac{1}{2}\beta\hbar\omega \equiv f(\beta)$$

$$P = \frac{1}{\beta}\frac{N_g}{V}$$

(5) $\frac{N}{V} = f(\beta_c) \simeq \left(\frac{m}{2\pi\beta_c\hbar^2}\right)^{3/2}\beta_c\hbar\omega = \left(\frac{m}{2\pi}\right)^{3/2}\frac{\omega}{\hbar}\beta_c^{-\frac{1}{2}}$

$$\therefore k_B T_c = \frac{1}{\beta_c} = \left[\frac{N}{V}\hbar\left(\frac{m}{2\pi}\right)^{3/2}\right]^2 = \left(\frac{N}{V}\right)^2\frac{\hbar^2}{\omega^2}\left(\frac{m}{2\pi}\right)^3$$

45. 状態密度 $D(\varepsilon) = \tau \varepsilon^{\frac{1}{2}}$

(1) $E = \int_0^\infty d\varepsilon\, D(\varepsilon) \frac{\varepsilon}{e^{\beta(\varepsilon-\mu)}+1}$ (Fermion) $\qquad N = \int_0^\infty D(\varepsilon) \frac{1}{e^{\beta(\varepsilon-\mu)}+1}$

$= \tau \beta^{-\frac{5}{2}} F_{\frac{3}{2}}(\alpha)\, \Gamma\left(\frac{3}{2}+1\right) \qquad = \tau \beta^{-\frac{3}{2}} F_{\frac{1}{2}}(\alpha)\, \Gamma\left(\frac{3}{2}\right)$

$N_A \frac{E}{N} = \frac{N_A}{\beta} \frac{3}{2} \frac{F_{\frac{3}{2}}(\alpha)}{F_{\frac{1}{2}}(\alpha)}$ $\qquad F = \mu N + \Omega = \mu N - PV$

$\| \frac{2}{3} E$

$S = -\frac{\partial \Omega}{\partial E} \qquad \Omega = \frac{1}{\beta} \int_0^\infty d\varepsilon\, D(\varepsilon) \log\left(1+e^{\beta(\mu-\varepsilon)}\right)$

$S = k\beta^2 \frac{\partial}{\partial \beta}\left[\frac{1}{\beta}\int_0^\infty d\varepsilon\, D(\varepsilon) \log\left(1+e^{\beta(\mu-\varepsilon)}\right)\right]$

$= k \int_0^\infty d\varepsilon\, D(\varepsilon) \log\left(1+e^{\beta(\mu-\varepsilon)}\right) - k\beta \int_0^\infty d\varepsilon\, \frac{D(\varepsilon)(\mu-\varepsilon)}{1+e^{\beta(\mu-\varepsilon)}}$

$= \frac{3}{2} k \frac{T}{\beta^{\frac{3}{2}}} F_{\frac{3}{2}}(\alpha) - \frac{k\tau}{\beta^{\frac{3}{2}}}\left(\alpha F_{\frac{1}{2}}(\alpha)\Gamma\left(\frac{3}{2}\right) - F_{\frac{3}{2}}(\alpha)\Gamma\left(\frac{5}{2}\right)\right)$

$N\frac{S}{N} = \left(\frac{3}{2}+1\right) R N_A \frac{F_{\frac{3}{2}}(\alpha)}{F_{\frac{1}{2}}(\alpha)} - \alpha k N_A$

$C_V = T\frac{\partial S}{\partial T} = -\beta \frac{\partial S}{\partial \beta} = -\alpha \frac{\partial S}{\partial \alpha} = -\frac{5}{2} R\alpha \frac{\partial}{\partial \alpha}\left(\frac{F_{\frac{3}{2}}(\alpha)}{F_{\frac{1}{2}}(\alpha)}\right) + \alpha R$

$= -\frac{5}{2} R\alpha \left\{\frac{F'_{\frac{3}{2}}(\alpha) F_{\frac{1}{2}}(\alpha) - F'_{\frac{1}{2}}(\alpha) F_{\frac{3}{2}}(\alpha)}{F_{\frac{1}{2}}(\alpha)^2}\right\} + \alpha R$

$F'_s(\alpha) = \frac{1}{\Gamma(s+1)} \int_0^\infty d\alpha\, \frac{x^s e^{x-\alpha}}{(e^{x-\alpha}+1)^2} = \frac{1}{\Gamma(s)} \int_0^\infty d\alpha\, \frac{x^{s-1}}{e^{x-\alpha}+1} = F_{s-1}(\alpha)$

$C_V = -\frac{3}{2} R\alpha + \frac{5}{2}\alpha R \frac{F'_{\frac{1}{2}}(\alpha) F_{\frac{3}{2}}(\alpha)}{F_{\frac{1}{2}}(\alpha)^2}$

(2)

45 (2)

$E(k) = \frac{\hbar^2 k^2}{2m}$ で大きな体積で周期的境界条件をほどこして $k^2 = \left(\frac{2\pi}{L}\right)^2 n^2$

波数空間内で同じエネルギーをもつ状態数(密度)は $d^3n = \frac{V}{(2\pi)^3} d^3k$

$$\int_{\varepsilon+d\varepsilon} d^3n = \frac{4\pi V}{(2\pi)^3} k^2 dk = \frac{2\cdot 4\pi V}{(2\pi)^3} \left(\frac{2m}{\hbar^2}\right)^{\frac{3}{2}} \frac{\varepsilon d\varepsilon}{2\varepsilon^2}$$

$$\therefore D(\varepsilon) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{\frac{3}{2}} \varepsilon^{\frac{1}{2}}$$

$\varepsilon = \frac{p^2}{2m}$

$2m\varepsilon = p^2$

$2m\, d\varepsilon = 2p\, dp$

$T=0$ で $N = \int_0^{\mu_0} D(\varepsilon) d\varepsilon = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{\frac{3}{2}} \frac{2}{3} \mu_0^{\frac{3}{2}}$

$$\frac{N}{V} = \frac{1}{3\pi^2} \left(\frac{2m}{\hbar^2}\right)^{\frac{3}{2}} \frac{3}{2} \mu_0^{\frac{3}{2}}$$

$\frac{dN}{dT} = 0$ より $\frac{\partial \alpha}{\partial T}$ をもとめればよいので

$N = \gamma T^{\frac{3}{2}} F_{\frac{1}{2}}(\alpha)$

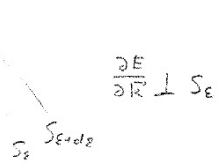
$$\frac{dN}{dT} = \frac{3}{2} \gamma T^{\frac{1}{2}} F_{\frac{1}{2}}(\alpha) + \gamma T^{\frac{3}{2}} \frac{\partial \alpha}{\partial T} \frac{\partial}{\partial \alpha} F_{\frac{1}{2}}(\alpha) = 0$$

$$\frac{\partial \alpha}{\partial T} = -\frac{3}{2T} \frac{F_{\frac{1}{2}}(\alpha)}{F'_{\frac{1}{2}}(\alpha)}$$

$$T\frac{\partial S}{\partial T} = T \frac{\partial \alpha}{\partial T} \frac{\partial S}{\partial \alpha}$$

$$= T \frac{\partial \alpha}{\partial T} \left(\frac{3}{2} R - \frac{5}{2} \frac{F_{\frac{3}{2}} F'_{\frac{1}{2}}}{F_{\frac{1}{2}}^2} \right)$$

46. (1) $E(\vec{k}) = \varepsilon$ 인 $\vec{k}$ 공간 상의 S_ε 와 $\frac{\partial E}{\partial \vec{k}} \perp S_\varepsilon$



$$D(\varepsilon)\,d\varepsilon = \frac{2V}{(2\pi)^3}\int_{\varepsilon \le E(\vec{k}) \le \varepsilon+d\varepsilon} d\vec{k}$$

$$= \frac{2V}{(2\pi)^3}\int_{S_\varepsilon} dS \cdot \frac{d\varepsilon}{\left|\frac{\partial E(\vec{k})}{\partial \vec{k}}\right|}$$

$$D(\varepsilon) = \frac{2V}{(2\pi)^3}\int_{S_\varepsilon} \frac{dS}{\left|\frac{\partial E}{\partial \vec{k}}\right|}$$

$$f(\varepsilon) = -\frac{\partial \Omega}{\partial \varepsilon} = \frac{\partial \Omega}{\partial \varepsilon} = \frac{1}{e^{\beta(\varepsilon-\mu)} \pm 1}$$

$$\Omega(\varepsilon) = \mp \frac{1}{\beta} \ln\left[1 \pm e^{\beta(\mu-\varepsilon)}\right]$$

$$\frac{dN(\varepsilon)}{d\varepsilon} = D(\varepsilon) \qquad PV = -\int d\varepsilon\, \Omega(\varepsilon) d\varepsilon$$

(2) $$PV = -\left[\,[N(\varepsilon)\Omega(\varepsilon)]_0^\infty - \int_0^\infty d\varepsilon\, f(\varepsilon)\Omega(\varepsilon)\,\right]$$

$$= \int d\varepsilon\, f(\varepsilon)\,\Omega(\varepsilon) \qquad D(\varepsilon) \propto \varepsilon^\delta \text{ 꼴}$$

$$\frac{PV}{E} = \frac{1}{\delta+1}$$

$$E = \int d\varepsilon\, f(\varepsilon)\, D(\varepsilon)\, \varepsilon$$

51. $Z = \sum e^{\beta \sum \mu H \sigma_i + J \sigma_i \sigma_{i+1}}$

$$V_{i,i+1} = J\sigma_i\sigma_{i+1} + \frac{1}{2}\mu H(\sigma_i + \sigma_{i+1})$$

$$A = e^{\beta V} = \begin{pmatrix} e^{\beta\mu H + \beta J} & e^{-\beta J} \\ e^{-\beta J} & e^{-\beta\mu H + \beta J} \end{pmatrix}$$

(2) 固有値 $\lambda = e^{\beta J}\left(\cosh \beta\mu H \pm \sqrt{\sinh^2 \beta\mu H + e^{-4\beta J}}\right)$

$$Z = \mathrm{Tr}\, A^N \sim \lambda_+^N = e^{N\beta J}\left\{\cosh \beta\mu H + \sqrt{\sinh^2 \beta\mu H + e^{-4\beta J}}\right\}^N$$

(1) $H=0$; $\lambda_+ = e^{\beta J}(1 + e^{-2\beta J}) = 2\cosh \beta J$ $\quad \therefore Z = 2^N \cosh^N \beta J$

(2) Z は β の analytic fn

$$F = -\frac{1}{\beta} \ln Z = -\frac{1}{\beta}\left(N \ln 2 + N \ln \cosh \beta J\right)$$

$$C = \frac{\partial U}{\partial T} = \frac{1}{\partial T}\left(F + TS\right) = NJ \frac{1}{\cosh^2 \beta J}$$

$S = -\frac{\partial F}{\partial T}$

(3) $M = \sum_i \mu \langle \sigma_i \rangle = \frac{1}{\beta}\frac{\partial}{\partial H} \ln Z = N\mu \dfrac{\sinh \beta\mu H}{\sqrt{\quad} + \cosh \beta\mu H}$

$$\left[\cosh \beta\mu H \sqrt{\quad} + \sinh^2 \beta\mu H + e^{-4\beta J}\right]$$

$$\chi = \left.\frac{\partial M}{\partial H}\right|_{H=0} = \cdots = N\mu^2 \beta\, e^{2\beta J}$$

$-NJ \frac{1}{\cosh^2 \beta J}$ $\quad \frac{\partial\left(\frac{1}{kT^2}\right)}{\partial T} = -\frac{2}{kT^3}$

$\frac{1}{kT^2}$ $\quad -N\frac{2\sinh}{2\cosh} \cdot \beta J$

(2) $\frac{C_v - S^2}{C_v}$

52.

1:ケミ ↑↑↑↓↑↑

$\Delta E = 2\mu H + 4J \sim O(1)$

$\Delta S = k_B \ln N \sim O(\log N)$

$\Delta F = \Delta E - T\Delta S \sim O(\log N)$

< 0

2:ケミ ↑↑↑↑↑
↑↑↓↑↑
↑↑↑↑↑

$\log \frac{N^D!}{(N^2 - N^D)!\, N^D!}$

$\sim O(N^D \log N)$

$\Delta E = N(2\mu H + 4J) \sim O(N)$

$\Delta S = k_B \log N \sim O(\log N)$

$\Delta F = \Delta E - T\Delta S \sim O(N)$

> 0

N^D $(D < 2)$

53.

(1) $Z = \sum_{\sigma_1=\pm1} \cdots \sum_{\sigma_N=\pm1} e^{\beta(J_z\langle\sigma\rangle + \mu H)\sigma_i}$

$= \left\{ 2\cosh \beta (J_z \langle\sigma\rangle + \mu H) \right\}^N$

$J_z\langle\sigma\rangle = \mu H'$ $\quad H' = \frac{J_z}{\mu}\langle\sigma\rangle$ 平均場近似

$H_{eff} = H + H'$

$M = \frac{kT}{V}\frac{\partial}{\partial H_{eff}} \log Z = \frac{N\mu}{V} \tanh \beta\mu \left(\frac{J_z\langle\sigma\rangle}{\mu} + H \right)$

$\therefore\ M = \frac{N}{V}\mu\langle\sigma\rangle$ $\quad \therefore \langle\sigma\rangle = \frac{VM}{N\mu} \equiv X$ $\quad X = \tanh(\beta J_z X + \beta\mu H)$

(2) $H=0$ のとき $X = \tanh \beta J_z X$ の解をもつ条件



$T < \frac{J_z}{k}$

(3) $\chi = \lim_{H\to 0} \frac{\partial M}{\partial H}$

$\frac{\partial M}{\partial H} = \frac{\partial}{\partial H}\left(N\mu \tanh \beta\mu \left(\frac{J_z V}{N\mu^2} M + H \right) \right) = \frac{N\mu \cdot \beta\mu \left(\frac{J_z V}{N\mu^2}\frac{\partial M}{\partial H} + 1 \right)}{\cosh^2 \beta\mu \left(\frac{J_z V}{N\mu^2} M + H \right)}$

$(1 - \beta J_z)\chi = \frac{N\beta\mu^2}{V}$

$\frac{J_z}{k} = T_c$

$\chi = \frac{N\mu^2}{k(T - T_c)V}$

$$\left.\frac{\partial F}{\partial M}\right|_{T,H} = F_2(T)\cdot 2M + F_4(T)\cdot 4M^3 - H = 0 \quad \cdots ①$$

$$\left.\frac{\partial^2 F}{\partial M^2}\right|_{T,H} = 2F_2(T) + 12F_4(T)\cdot M^2 > 0 \quad \cdots ②$$

(1) (i) $T > T_c$ $H=0$ のとき $M=0$ より②から $F_2(T) > 0$

① より $H=0$, $F_2(T)>0$ のとき $M=0$ 以外の解を持たない条件は $F_4(T) > 0$

(ii) $T < T_c$ $M=0$ 以外の解をもつことから ①より $F_2(T)$, $F_4(T)$ は異符号となり②より

$2F_2(T) + F_4 \cdot 4M^2 = 0 \quad \rightarrow M^2 = -\frac{F_2}{2F_4} \quad \rightarrow 2F_2 - 6F_4 > 0 \quad \therefore F_2 < 0,\ F_4 > 0$

(2) $F(T,H,M) = F_0(T) + a(T-T_c)M^2 + bM^4 - HM + \cdots \quad (a, b > 0)$

(i) $T > T_c \quad -T\frac{\partial^2}{\partial T^2}F_0(T)$

(ii) $T < T_c \quad \frac{\partial F}{\partial M} = 0 \rightarrow M = \sqrt{\frac{a}{2b}(T_c - T)} \quad F = F_0(T) - \frac{a^2}{4b}(T_c - T)^2$

$$\left(= -T\left(\frac{\partial^2 F}{\partial T^2}\right)_{T_c} = \frac{a^2 T_c}{2b} - T\left(\frac{\partial^2 F_0(T)}{\partial T^2}\right)\right) //$$

(3) $\frac{\partial F}{\partial M} = 2a(T-T_c)M + 4bM^3 - H = 0 \quad 4bM^3 = H$

$$\frac{\partial M}{\partial H} = \frac{1}{2a(T-T_c) + 12bM^2} = \frac{1}{b(T-T_c)} \quad \left(\frac{\partial M}{\partial H}\right) = \frac{1}{12bM^2} = \frac{1}{12b\left(\frac{H}{4b}\right)^{2/3}}$$

$T = T_c$; $H \rightarrow 0 \quad M = \sqrt{\frac{a}{2b}(T_c - T)}$

$$\begin{cases} \mathrm{rot}\,E + \frac{1}{c}\dot{H} = 0 \\ \mathrm{rot}\,H - \frac{1}{c}\dot{E} = 0 \end{cases}$$

$$\begin{cases} E = -\frac{1}{c}\dot{A} + \nabla A_0 \\ H = \mathrm{rot}\,A \end{cases}$$

$A_\mu \to A_\mu + \partial_\mu \Lambda$ (gauge transform)

$A_0 = 0$, $\mathrm{div}\,A = 0$

$$\Delta A - \frac{1}{c^2}\ddot{A} = \Box A = 0 \quad - ① \quad (\text{光の波動方程式})$$

$$H = \frac{1}{8\pi}\int (E^2 + H^2)\, d^3x$$

$$= \frac{1}{8\pi}\int \left\{ \frac{1}{c^2}\dot{A}^2 + (\mathrm{rot}\,A)^2 \right\} d^3x$$

完全反射の境界条件

$z=0$ の壁の面で $\begin{cases} E \text{ の接線成分} \\ H \text{ の法線成分} \end{cases} = 0$

$E_x = E_y = 0 \to \dot{A}_x = \dot{A}_y = 0$

$H_z = 0 \to \partial_x A_y - \partial_y A_x = 0$

$$\left.\begin{aligned} A_x &= \partial_x \phi(x,y) \\ A_y &= \partial_y \phi(x,y) \end{aligned}\right\} \to A_x = A_y = 0$$

$$A_x = \sum_s Q_{s,x} \cos\frac{\pi s_x x}{L} \sin\frac{\pi s_y y}{L} \sin\frac{\pi s_z z}{L}$$

$$A_y = \sum_s Q_{s,y} \sin\frac{\pi s_x x}{L} \cos\frac{\pi s_y y}{L} \cdot \sin\frac{\pi s_z z}{L} \quad - ②$$

$$A_z = \sum_s Q_{s,z} \sin\frac{\pi s_x x}{L} \sin\frac{\pi s_y y}{L} \cos\frac{\pi s_z z}{L}$$

$s = (s_x\ s_y\ s_z)$: 正の整数

但し $Q_s \cdot \$ = 0$ (横波) $\quad \mathbb{Q}_s = q_s^{(1)} e_s^{(1)} + q_s^{(2)} e_s^{(2)} \quad [e_s^{(i)} \cdot \$ = 0]$

②を①に代入すると

$$\frac{\pi^2 s^2 c^2}{L^2} Q_s + \ddot{Q}_s = 0 \to \begin{cases} q_s^{(1)} = a_s^{(1)} \cos(2\pi \nu_s t + \delta_s^{(1)}) \\ q_s^{(2)} = a_s^{(2)} \cos(2\pi \nu_s t + \delta_s^{(2)}) \end{cases} \qquad \nu_s = \frac{s}{2L} c$$

No.

55

(1) 状態数 $W = e^{\frac{S}{k}}$ $\Delta S = \dfrac{\Delta E - T\Delta S + P\Delta V}{T}$

(2) $\Delta E = \underset{T}{\frac{\partial E}{\partial S}}\Delta S + \underset{-P}{\frac{\partial E}{\partial V}}\Delta V + \frac{1}{2}\left(\frac{\partial^2 E}{\partial S^2}(\Delta S)^2 + 2\frac{\partial^2 E}{\partial S \partial V}(\Delta S \Delta V) + \frac{\partial^2 E}{\partial V^2}(\Delta V)^2\right)$

$\frac{1}{2}\left(\Delta S\, \Delta\left(\underset{\Delta T}{\frac{\partial E}{\partial S}}\right)_V - \Delta V\, \Delta\left(\underset{\Delta P}{\frac{\partial E}{\partial V}}\right)_S\right)$

(3) $\Delta S = \left(\frac{\partial S}{\partial T}\right)_V \Delta T + \left(\frac{\partial S}{\partial V}\right)_T \Delta V = \frac{C_v}{T}\Delta T + \left(\frac{\partial P}{\partial T}\right)_V \Delta V$

$\Delta P = \left(\frac{\partial P}{\partial T}\right)_V \Delta T + \left(\frac{\partial P}{\partial V}\right)_T \Delta V$ $\chi_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$

$\int fg' = fg - \int f's$

$e^{-\frac{t}{\tau}}$ $\quad g = -\tau e^{-\frac{t}{\tau}}$

No. ( )

57

$$M\frac{dV}{dt} = -\frac{V}{B} + F(t)$$

$$\left(M\frac{d}{dt} + \frac{1}{B}\right)V = F(t) \qquad V(t) = e^{-\frac{t}{MB}}U(t)$$

$$Me^{-\frac{t}{MB}}\frac{dU}{dt} - \frac{1}{B}e^{-\frac{t}{MB}}U = -\frac{1}{B}e^{-\frac{t}{MB}}U + F(t)$$

$$\frac{dU}{dt} = \frac{1}{M}e^{\frac{t}{MB}}F(t) \qquad U = \int_0^t \frac{1}{M}e^{\frac{s}{MB}}F(s)\,ds$$

$$\therefore V(t) = \frac{1}{M}e^{-\frac{t}{MB}}\int_0^t e^{\frac{s}{MB}}F(s)\,ds + V(0)e^{-\frac{t}{MB}}$$

$$= V_0 e^{-\frac{t}{\tau}} + \frac{1}{M}\int_0^t e^{-\frac{t-s}{\tau}}F(s)\,ds$$

$$\overline{V(t)} = \lim_{T\to\infty}\frac{1}{T}\int_0^T V(t)\,dt = \lim_{T\to\infty}\frac{1}{T}\int_0^T\left\{V_0 e^{-\frac{t}{\tau}} + \frac{1}{M}\int_0^t e^{-\frac{t-s}{\tau}}F(s)\,ds\right\}dt$$

$$= \lim_{T\to\infty}\frac{1}{T}\int_0^T V_0 e^{-\frac{t}{\tau}}dt + \frac{1}{M}\lim_{T\to\infty}\frac{1}{T}\int_0^T e^{-\frac{t}{\tau}}\int_0^t e^{\frac{s}{\tau}}F(s)\,ds\,dt$$

$\to 0$ $\qquad$ $f(t)$

$$-\tau e^{-\frac{t}{\tau}}f(t)\Big|_0^T + \int_0^T \tau e^{-\frac{t}{\tau}}f'(t)\,dt$$

$$= \frac{1}{M}\lim_{T\to\infty}\frac{1}{T}\left[-\tau e^{-\frac{T}{\tau}}\int_0^T e^{\frac{s}{\tau}}F(s)\,ds + \int_0^T \tau e^{-\frac{t}{\tau}}e^{\frac{t}{\tau}}F(t)\,dt\right]$$

$\to 0$ Finite $\qquad$ $\overline{F(t)} = 0$

$\langle x - \langle x\rangle\rangle^2 = \langle x^2\rangle - \langle x\rangle^2$